LIPSedge™ L210u / L215u
High-Res Structured-Light Camera

User's Manual

Aug 2024

Revision 1.2.1

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December 2023

LIPS
Sense The World

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Revision History

Revision	Description	Date
1.2.1	Modified the naming of Samples from pages 53 to 59.	2024/08/01
1.2	Add the Linux x64& Jetson arm64 installation guide in Chapter3 3.1-B & Chapter3 3.1-C.	2024/04/19
1.1	Re-Modified the User Guide Frame	2024/03/08
1.0	New Release	2023/10/24

1. Package & Hardware Overview

1.1 Package Overview

A. Packing List

This section applies to LIPSedge™ L210u/ L215u ONLY. The packing list serves as a reference for package contents. If anything is missing, contact info@lips-hci.com. For technical specifications, refer to *LIPSedge™ L210u/ L215u datasheet*.

No.	Item	Qty.
1.	LIPSedge™ L210u / L215u High-Res Structured-Light Camera	1
2.	LIPSedge™ L210u / L215u 3D High-Res Structured-Light Camera Quick Start Guide	1
3.	USB 2.0 Y-Cable	1

B. Camera Accessories

The following supplementary options facilitate the installation process but are NOT included in the package. Prepare the following accessories before proceeding to the installation process:

- Camera tripod: Secures the camera to a stable position.
- PC / laptop: LIPSedge™ L Series operates on PC / laptops with 7th Generation 7th Generation Intel® Core™ i5 Processors.

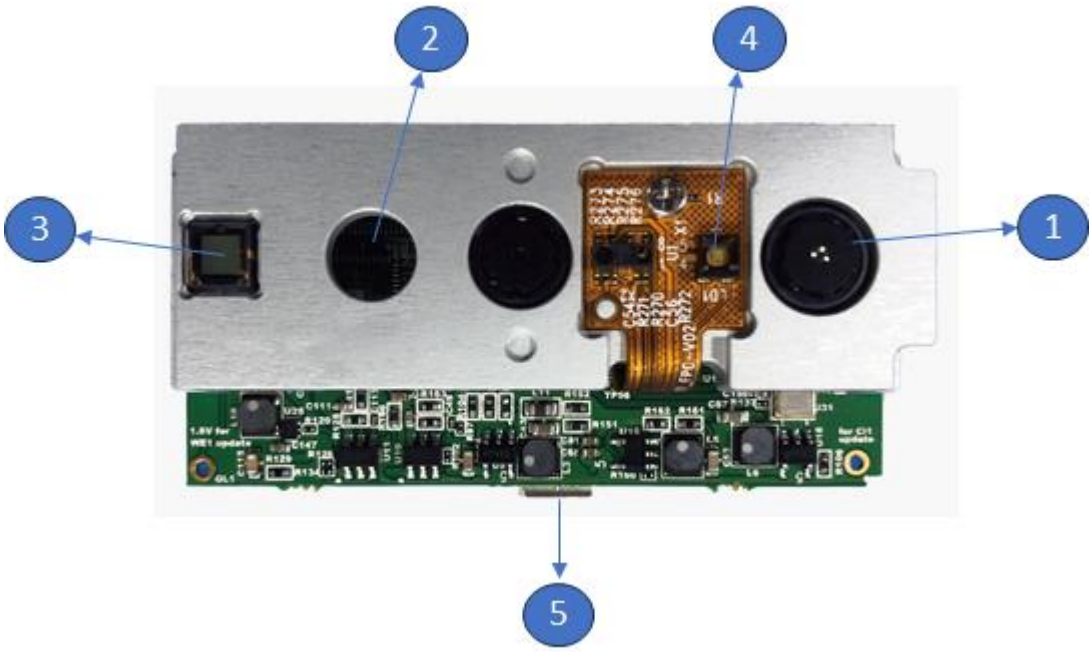
C. Supplementary Options

This section is NOT applicable to this model.

1.2 Hardware Overview

A. Hardware Features

LIPSedge™ L210u



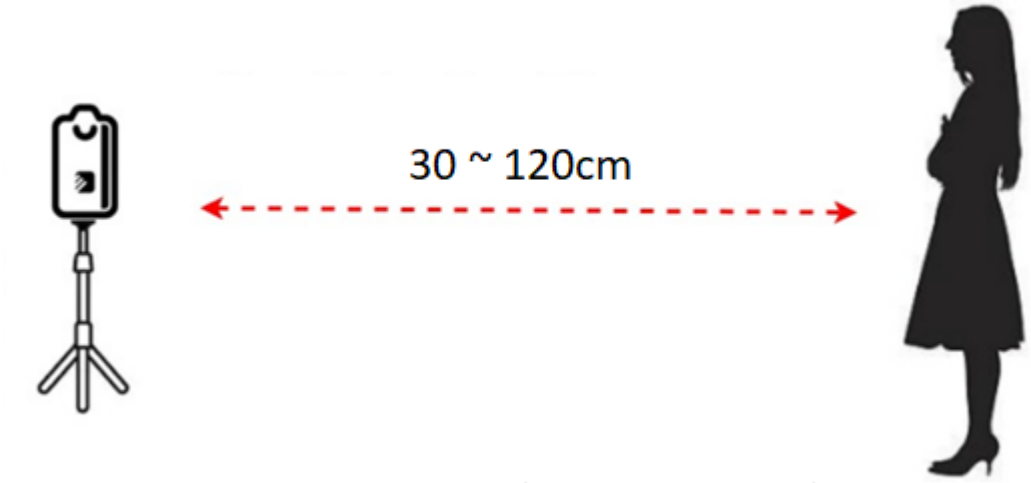
No.	Key Component	Description
1.	NIR sensor	Receives the IR image
2.	RGB sensor	Receives the RGB image
3.	NIR illuminator	Projects NIR structured dots light
4.	Flood Projector	Emits uniform, flat IR light to enhance lightness
5.	Micro USB Port	Provides power input and data output to the host PC / laptop
Note: Do NOT obscure or block the openings of the camera's optical components to ensure the optimal performance of the camera.		

LIPSedge™ L215u

No.	Key Component	Description
1.	NIR sensor	Receives the IR image
2.	RGB sensor	Receives the RGB image
3.	NIR illuminator	Projects NIR structured dots light
4.	Flood Projector	Emits uniform, flat IR light to enhance lightness
5.	Micro USB Port	Provides power input and data output to the host PC / laptop
6.	M3 * 4 mm * 2	Major mount holes
7.	Tripod mount	Mounts the camera onto the tripod
Note: Do NOT obscure or block the openings of the camera's optical components to ensure the optimal performance of the camera.		

B. Hardware Installation

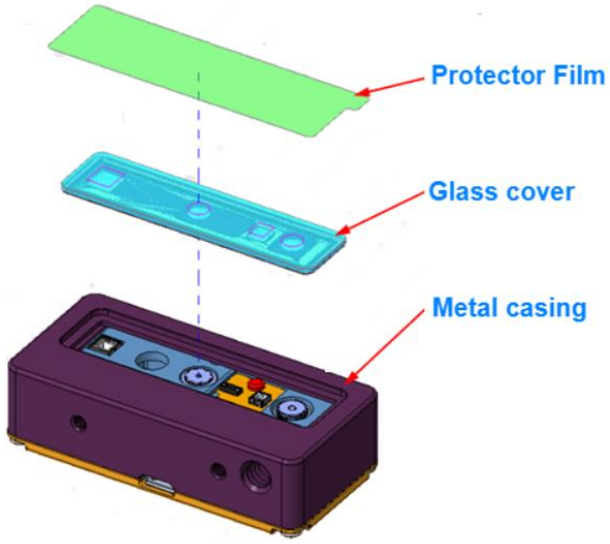
For optimal recognition results, follow these requirements when installing your LIPS camera.



- **Working Distance**: Install the camera approximately **40 - 120 cm** from the recognition target, depending on the working range for different scenarios. To adjust the camera's working range, refer to *Working Range Settings*.
- **Camera Position**: Install the camera on a **tripod** facing the recognition target. Note that LIPS Corp. does NOT provide a tripod within the package, and you must get your tripod for hardware installation.
- **Illumination**: AVOID installing the camera at places under **direct sunlight** or with an illumination level **above 1000 lux**.

C. Camera Connection

WARNING: A thin layer of film is applied to the camera lens for protective purposes when shipped. Remove the film ONLY when you are ready to use the camera. Keeping the thin film while streaming may cause the image quality to decrease.



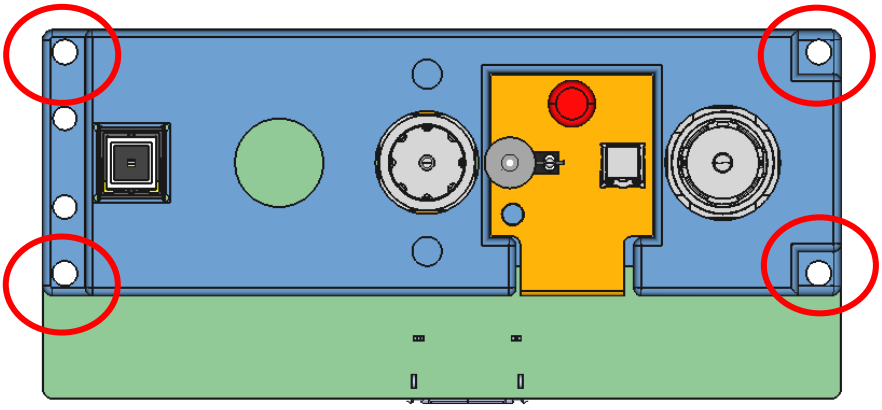
1. Plug the Micro B end of the USB-Y cable to the LIPS Camera.

2. Plug both **Type-A** end of the USB-Y cable to your **PC / laptop**.

D. Mounting Options

LIPSedge™ L Series camera supports a bracket mount, enabling you to secure the camera to a load-bearing structure, such as a wall or a pole surface.

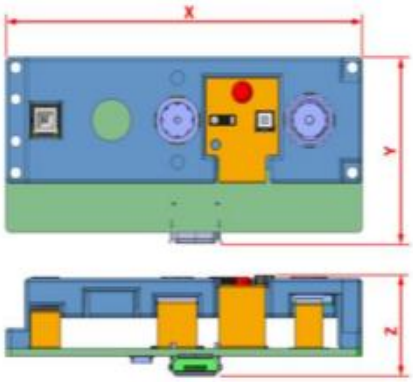
Once the SLiM protection is in place, secure the bracket mount to the desired load-bearing structure using four M16 screws inserted through the four Ø 1.7 fixing holes illustrated in the figure below:



Note: Do NOT obscure or block the openings of the camera's optical components to ensure the optimal performance of the camera.

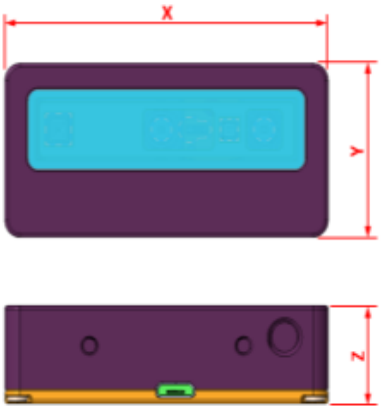
E. Hardware Dimensions

LIPSedge™ L210u



Dimension	MIN	NOM	MAX	TOLERANCE	UNIT
X	53.8	54	54.2	±0.2	mm
Y	26.3	26.6	26.9	±0.3	mm
Z	13.9	14.3	14.7	±0.4	mm

LIPSedge™ L215u

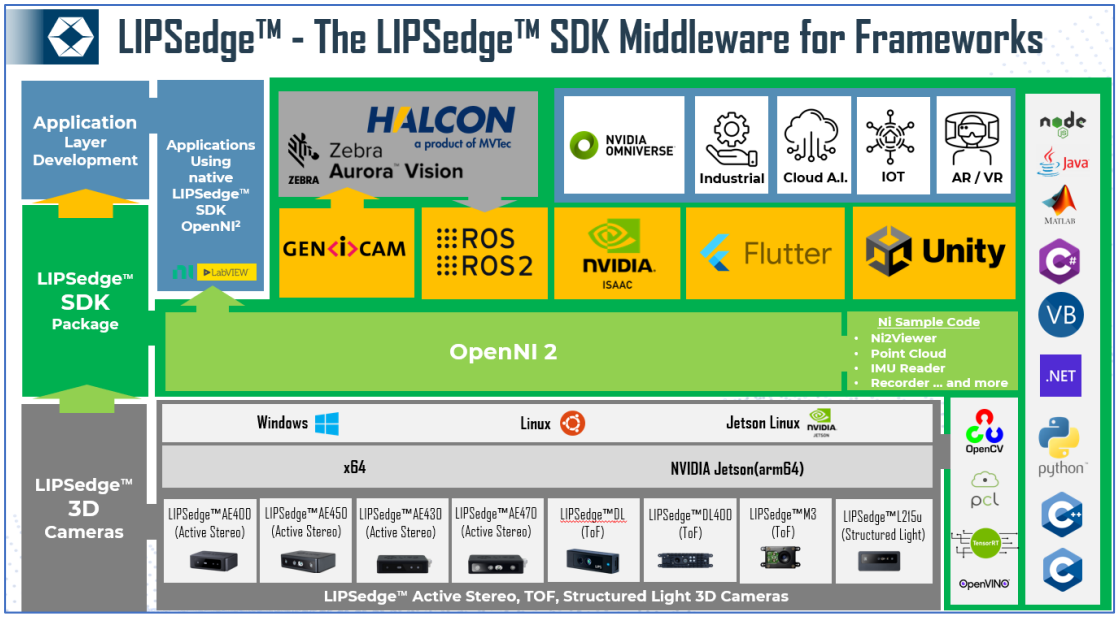


Dimension	MIN	NOM	MAX	TOLERANCE	UNIT
X	57.7	63	63.3	±0.3	mm
Y	29.9	30.2	30.5	±0.3	mm
Z	16.7	17	17.3	±0.3	mm

2. SDK Structure

LIPS 3D camera / SDK offers a system for developing depth-sensing applications. As the LIPS system architecture illustrates, the system is comprised of the **hardware layer** and the **software layer**.

The hardware layer oversees data capture, transfer, and processes. In the software layer, the captured data is fetched by the LIPS SDK (Software Development Kit) on the OS environment. Depending on the project complexity, wrappers and third-party utilities may be engaged before the data is eventually presented in the application layer for business applications.



The core of the system, the LIPS SDK, is comparable to a toolbox full of software modules comprised of middleware, libraries, wrappers and API, and miscellaneous programming languages / platforms for application development. With extensive wrapper support, LIPS SDK enables developers to access bottom layer data with APIs, thus eliminating the hassle of changing third-party functions. The result is a highly effective project scoping, monitoring, and execution workflow compatible with the fast-pacing AIoT market and machine vision demands.

3. SDK Installation

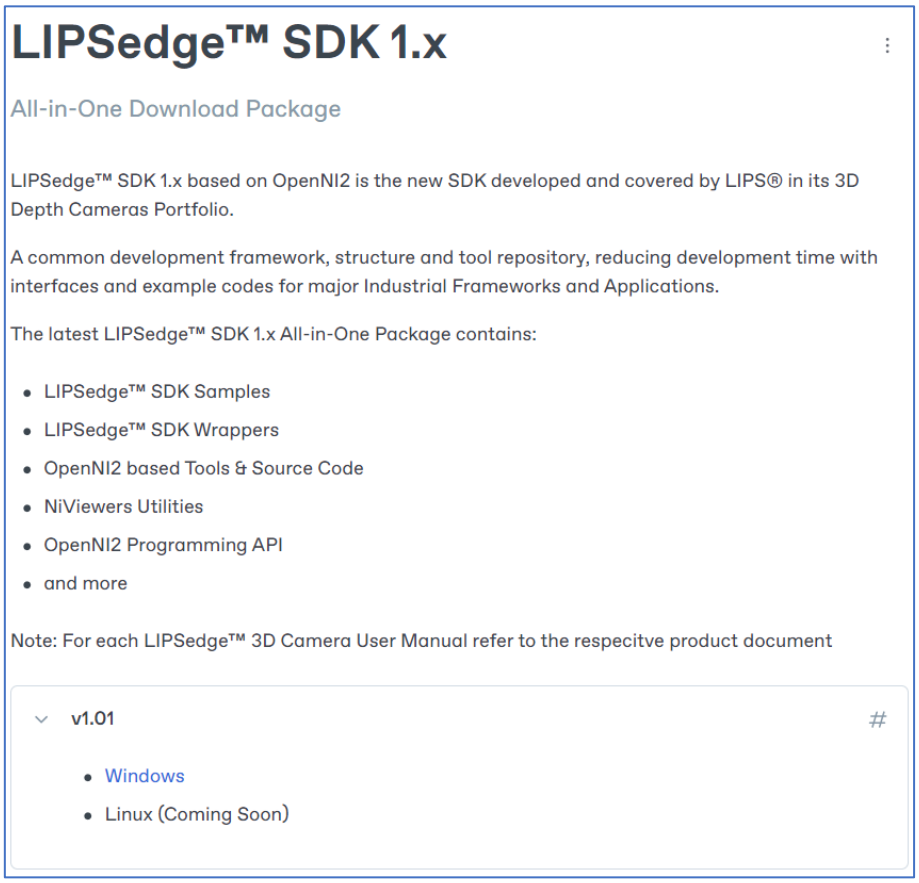
3.1 Platforms

A. Windows Installation (x64)

a. SDK Installation

To access the SDK, visit [3D Depth Camera SDK Free Download | LIPSedge™ SDK (lips-hci.com)](#) and procure the package for the preferred version, framework, and compatible operating system.

1. Visit [LIPSedge™ SDK](#) website.



LIPSedge™ SDK 1.x

All-in-One Download Package

LIPSedge™ SDK 1.x based on OpenNI2 is the new SDK developed and covered by LIPS® in its 3D Depth Cameras Portfolio.

A common development framework, structure and tool repository, reducing development time with interfaces and example codes for major Industrial Frameworks and Applications.

The latest LIPSedge™ SDK 1.x All-in-One Package contains:

- LIPSedge™ SDK Samples
- LIPSedge™ SDK Wrappers
- OpenNI2 based Tools & Source Code
- NiViewers Utilities
- OpenNI2 Programming API
- and more

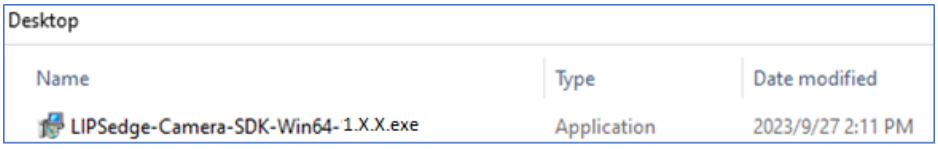
Note: For each LIPSedge™ 3D Camera User Manual refer to the respecitve product document

v1.01 #

- Windows
- Linux (Coming Soon)

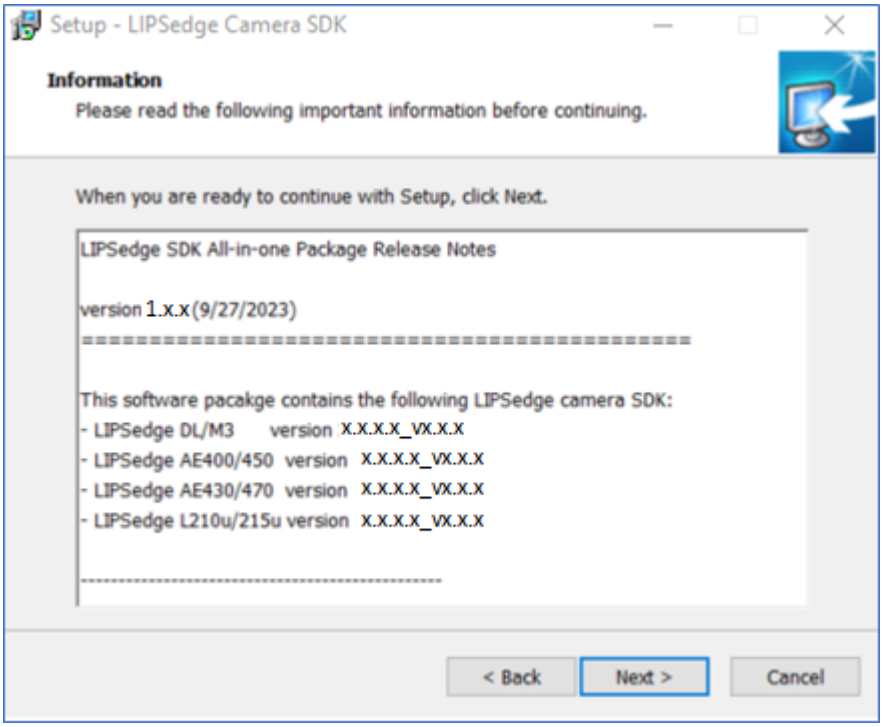
2. Under **LIPSedge™ Camera SDK**, select SDK file to download the installation file based on the preferred version, framework, and compatible operating system.

3. Extract and execute the downloaded file.

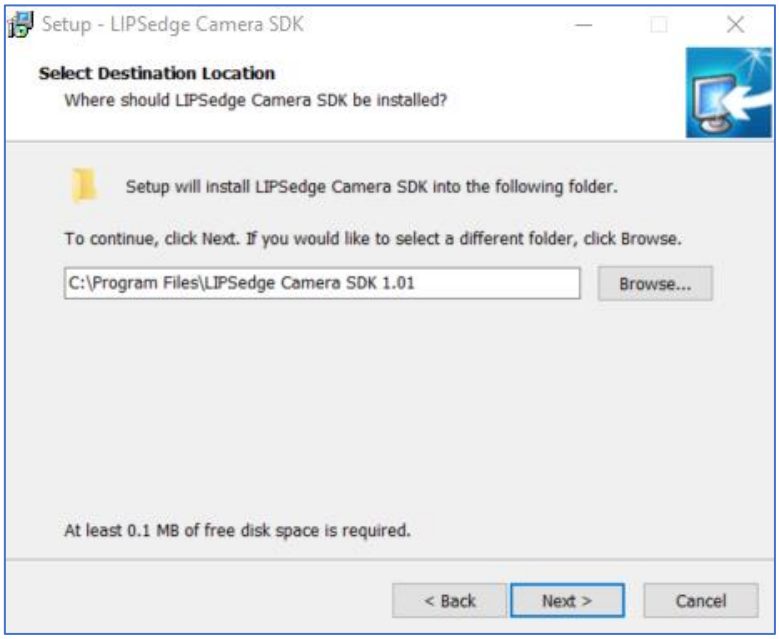


4. Click **Next**.

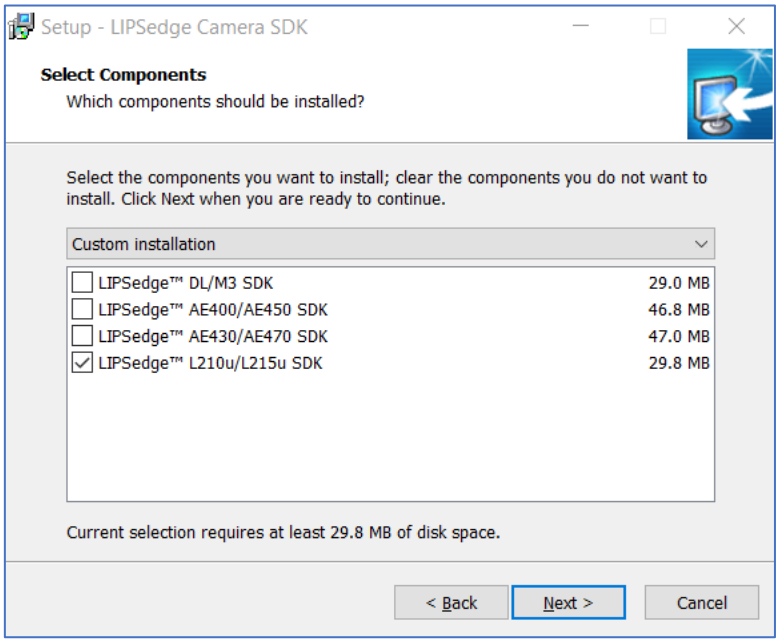
5. Click **Next**.



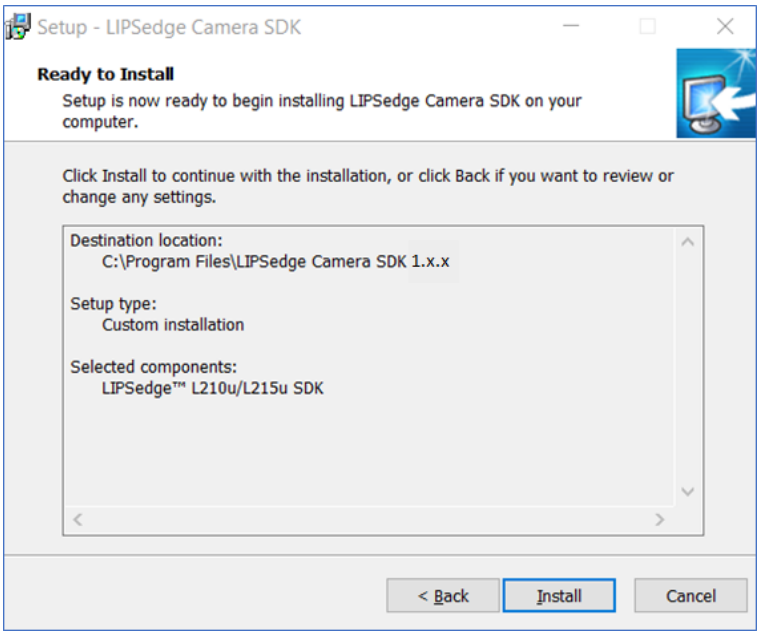
6. Select **the SDK installation location** and click **Next**.



7. From the top menu, select the type of installation for the host PC: In this case, it is recommended to select **Custom Installation > LIPSedge™ L210u / L215u SDK**.
   - **Full Installation**: Install all LIPSedge™ SDKs with all system dependencies for all available LIPSedge™ camera models.
   - **Compact Installations**: Install LIPSedge™ SDK with minimum system dependencies for the chosen model.
   - **Custom Installation**: Install LIPSedge™ SDK with limited functionality for the chosen model.

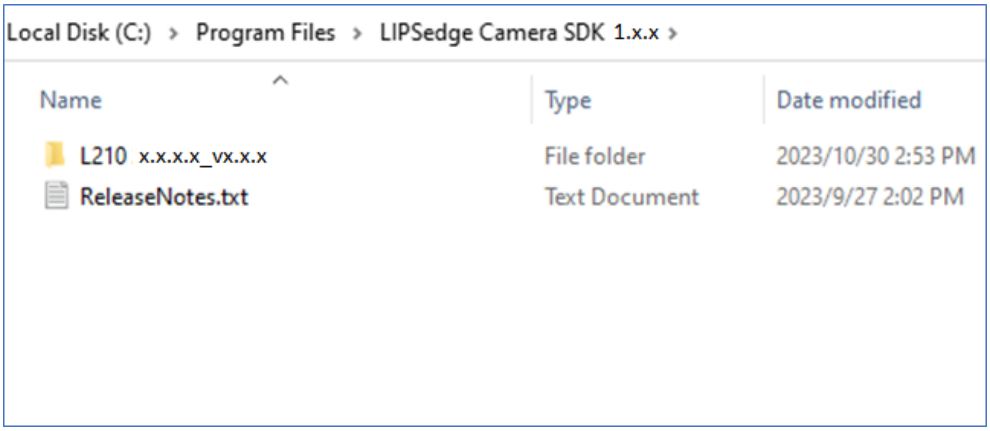


8. Click **Install**.



9. Click **Finish**.

10. Once the installation is successful, the LIPSedge™ SDK and its relevant components are accessible at C:\Program Files\LIPSedge Camera SDK 1.x.x.



11. The LIPSedge™ SDK is also accessible from the desktop shortcut.

12. The desktop shortcut folder contains:

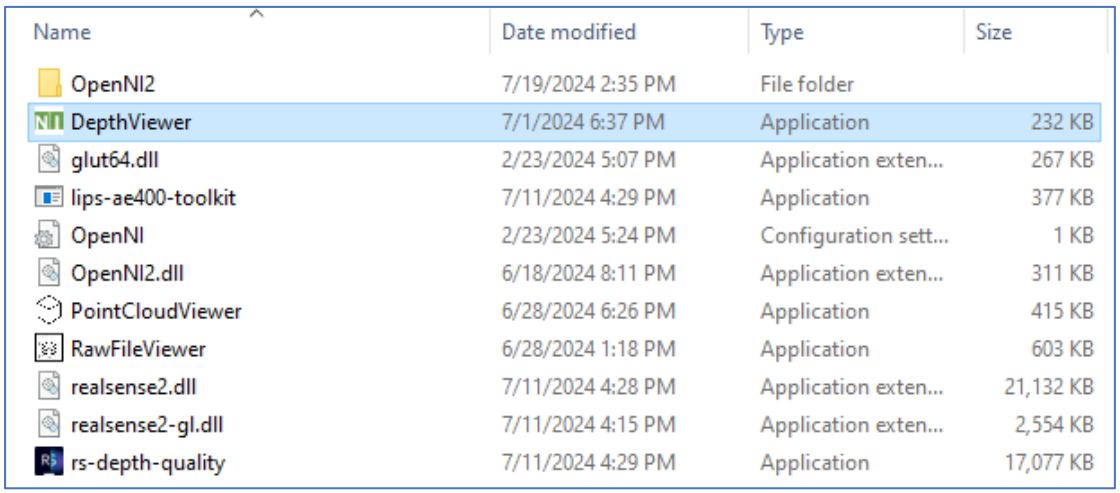
Name	Date modified	Type	Size
Browse LIPSedge SDK Samples	7/19/2024 2:35 PM	Internet Shortcut	1 KB
Browse LIPSedge SDK Wrappers	7/19/2024 2:35 PM	Internet Shortcut	1 KB
DepthViewer	7/19/2024 2:35 PM	Shortcut	2 KB
Download latest LIPSedge SDK	7/19/2024 2:35 PM	Internet Shortcut	1 KB
LIPS tools source code	7/19/2024 2:35 PM	Shortcut	2 KB
lips-hci.com	7/19/2024 2:35 PM	Internet Shortcut	1 KB
OpenNI2 Programming API	7/19/2024 2:35 PM	Shortcut	2 KB
PointCloudViewer	7/19/2024 2:35 PM	Shortcut	2 KB
Release Notes	7/19/2024 2:35 PM	Shortcut	2 KB
Run LIPS tools	7/19/2024 2:35 PM	Shortcut	3 KB
Uninstall LIPSedge DL SDK Vx.x.x	7/26/2024 10:22 AM	Shortcut	2 KB

- **Browse LIPSedge SDK Samples**: A link to LIPS Corp.'s GitHub, for example applications
- **Browse LIPSedge SDK Wrappers**: Link to LIPS Corp.'s GitHub for the supported wrapper's source code.
- **Download Latest SDK**: Link to LIPS Corp's official website for the latest SDK
- **LIPS tools source code**: Contains a series of executable tools and the source code of these tools.
- **lips-hci.com**: Link to LIPS Corp's official website.
- **DepthViewer**: An image viewer for streaming RGB / depth images. Refer to *Chapter3 3.1-A-c*.
- **OpenNI2 Programming API**: A link to OpenNI API's programming guide.
- **Release Notes**: Announcement of the latest feature.
- **Run LIPS tools**: Contains a series of executable tools.
- **Uninstall LIPSedge™ SDK**: A quick uninstallation link for LIPSedge™ SDK.

b. Camera Access

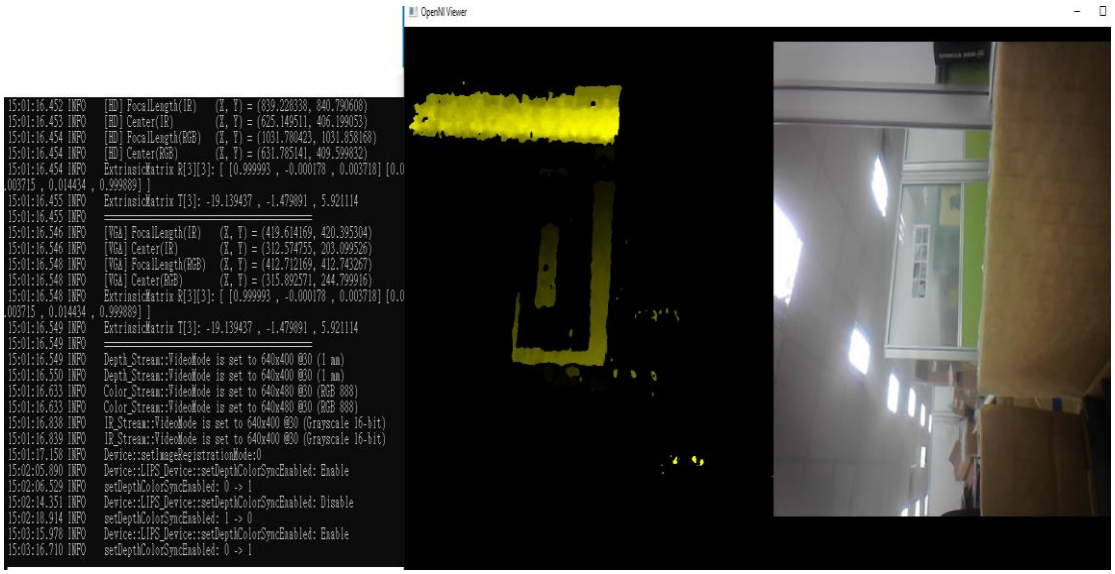
The LIPSedge™ L210u / L215u SDK (OpenNI based) enables the LIPSedge™ L210u / L215u camera to stream live 3D, RGB, and point-cloud images efficiently. The SDK supports a **single stream.**

1. Go to **LIPSedge™ AE400_AE450 SDK** > **OpenNI2** > **Tools** and start **DepthViewer**.

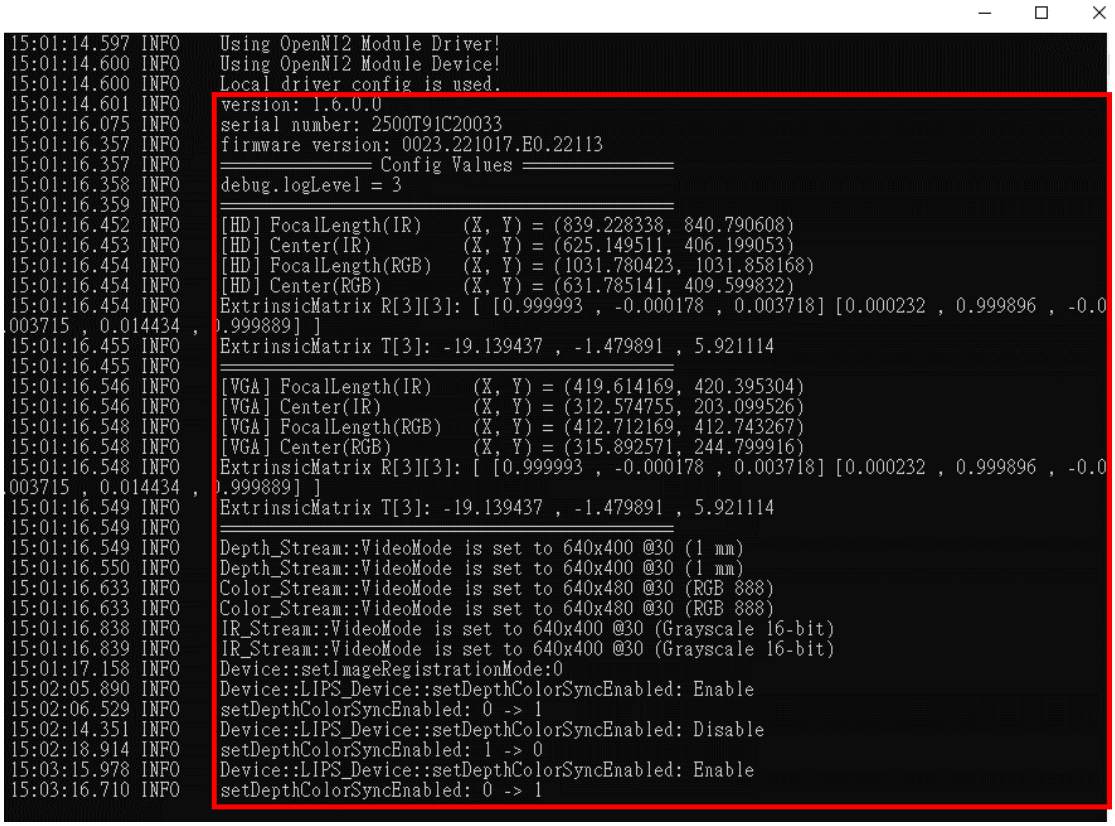


Name	Date modified	Type	Size
OpenNI2	7/19/2024 2:35 PM	File folder	
DepthViewer	7/1/2024 6:37 PM	Application	232 KB
glut64.dll	2/23/2024 5:07 PM	Application exten...	267 KB
lips-ae400-toolkit	7/11/2024 4:29 PM	Application	377 KB
OpenNI	2/23/2024 5:24 PM	Configuration sett...	1 KB
OpenNI2.dll	6/18/2024 8:11 PM	Application exten...	311 KB
PointCloudViewer	6/28/2024 6:26 PM	Application	415 KB
RawFileViewer	6/28/2024 1:18 PM	Application	603 KB
realsense2.dll	7/11/2024 4:28 PM	Application exten...	21,132 KB
realsense2-gl.dll	7/11/2024 4:15 PM	Application exten...	2,554 KB
rs-depth-quality	7/11/2024 4:29 PM	Application	17,077 KB

2. Upon activation, a **Status Window** and a **Viewer Window** both pop

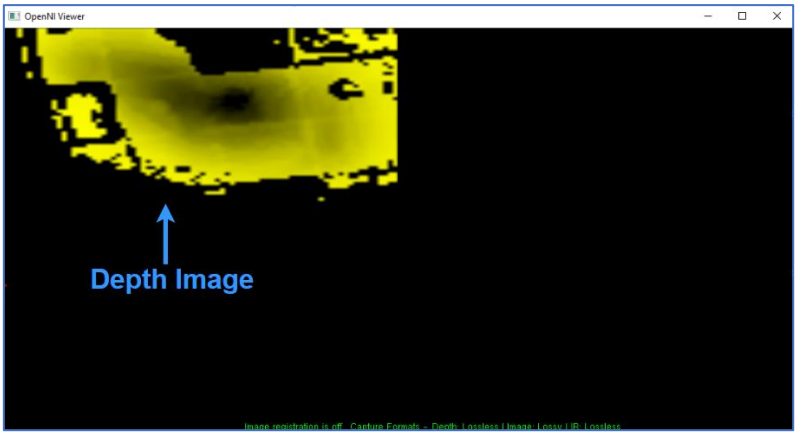


3. The **Status Window** displays the camera information, such as SDK build version, SDK time stamp, camera ID, and status.

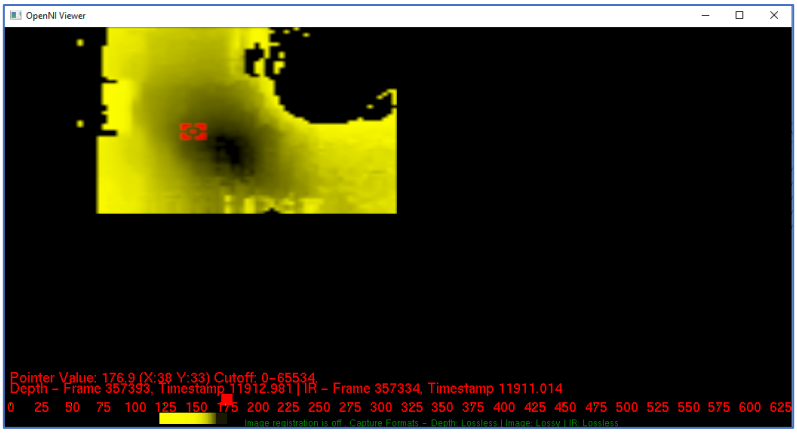


```
15:01:14.597 INFO    Using OpenNI2 Module Driver!
15:01:14.600 INFO    Using OpenNI2 Module Device!
15:01:14.600 INFO    Local driver config is used.
15:01:14.601 INFO    version: 1.6.0.0
15:01:16.075 INFO    serial number: 2500T91C20033
15:01:16.357 INFO    firmware version: 0023.221017.E0.22113
15:01:16.357 INFO    ========== Config Values ==========
15:01:16.358 INFO    debug.logLevel = 3
15:01:16.359 INFO    ==================================
15:01:16.452 INFO    [HD] FocalLength(IR)    (X, Y) = (839.228338, 840.790608)
15:01:16.453 INFO    [HD] Center(IR)         (X, Y) = (625.149511, 406.199053)
15:01:16.454 INFO    [HD] FocalLength(RGB)   (X, Y) = (1031.780423, 1031.858168)
15:01:16.454 INFO    [HD] Center(RGB)        (X, Y) = (631.785141, 409.599832)
15:01:16.454 INFO    ExtrinsicMatrix R[3][3]: [ [0.999993 , -0.000178 , 0.003718] [0.000232 , 0.999896 , -0.0
003715 , 0.014434 , 0.999889] ]
15:01:16.455 INFO    ExtrinsicMatrix T[3]: -19.139437 , -1.479891 , 5.921114
15:01:16.455 INFO    ==================================
15:01:16.546 INFO    [VGA] FocalLength(IR)    (X, Y) = (419.614169, 420.395304)
15:01:16.546 INFO    [VGA] Center(IR)         (X, Y) = (312.574755, 203.099526)
15:01:16.548 INFO    [VGA] FocalLength(RGB)   (X, Y) = (412.712169, 412.743267)
15:01:16.548 INFO    [VGA] Center(RGB)        (X, Y) = (315.892571, 244.799916)
15:01:16.548 INFO    ExtrinsicMatrix R[3][3]: [ [0.999993 , -0.000178 , 0.003718] [0.000232 , 0.999896 , -0.0
003715 , 0.014434 , 0.999889] ]
15:01:16.549 INFO    ExtrinsicMatrix T[3]: -19.139437 , -1.479891 , 5.921114
15:01:16.549 INFO    ==================================
15:01:16.549 INFO    Depth_Stream::VideoMode is set to 640x400 @30 (1 mm)
15:01:16.550 INFO    Depth_Stream::VideoMode is set to 640x400 @30 (1 mm)
15:01:16.633 INFO    Color_Stream::VideoMode is set to 640x480 @30 (RGB 888)
15:01:16.633 INFO    Color_Stream::VideoMode is set to 640x480 @30 (RGB 888)
15:01:16.838 INFO    IR_Stream::VideoMode is set to 640x400 @30 (Grayscale 16-bit)
15:01:16.839 INFO    IR_Stream::VideoMode is set to 640x400 @30 (Grayscale 16-bit)
15:01:17.158 INFO    Device::setImageRegistrationMode:0
15:02:05.890 INFO    Device::LIPS_Device::setDepthColorSyncEnabled: Enable
15:02:06.529 INFO    setDepthColorSyncEnabled: 0 -> 1
15:02:14.351 INFO    Device::LIPS_Device::setDepthColorSyncEnabled: Disable
15:02:18.914 INFO    setDepthColorSyncEnabled: 1 -> 0
15:03:15.978 INFO    Device::LIPS_Device::setDepthColorSyncEnabled: Enable
15:03:16.710 INFO    setDepthColorSyncEnabled: 0 -> 1
```

4. The **Viewer Window** displays the Depth image on the screen.



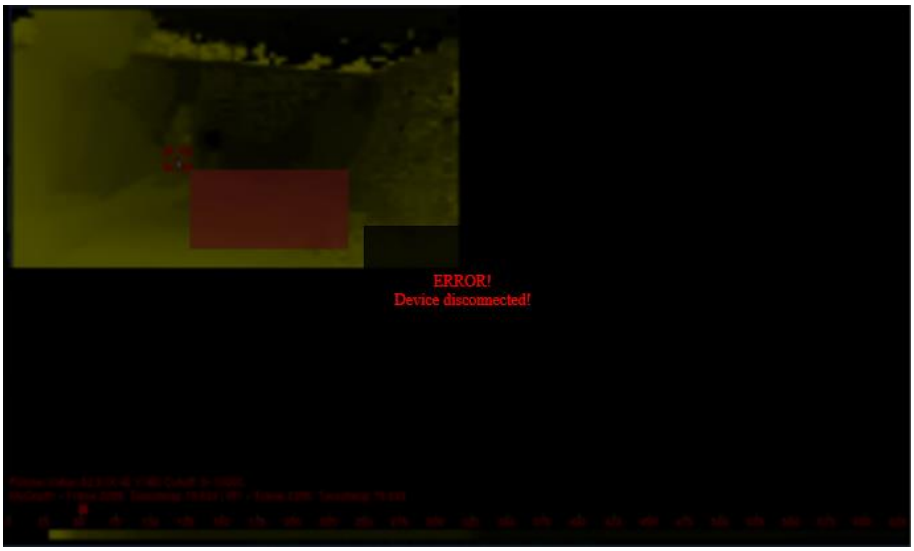
5. Press **[p]** to activate **the pointer mode**. The meter below shows the depth data of the given point marked by the indicator.



6. Measure the distance by dragging a mouse cursor from point1 to point2 as below.

7. Optionally press F to maximize/minimize the viewer's window.

Note: If, for any reason, the USB cable is forcibly removed while DepthViewer is functioning, an ERROR! The device disconnected message appears on the screen. To resume live streaming, re-plug the USB cable.



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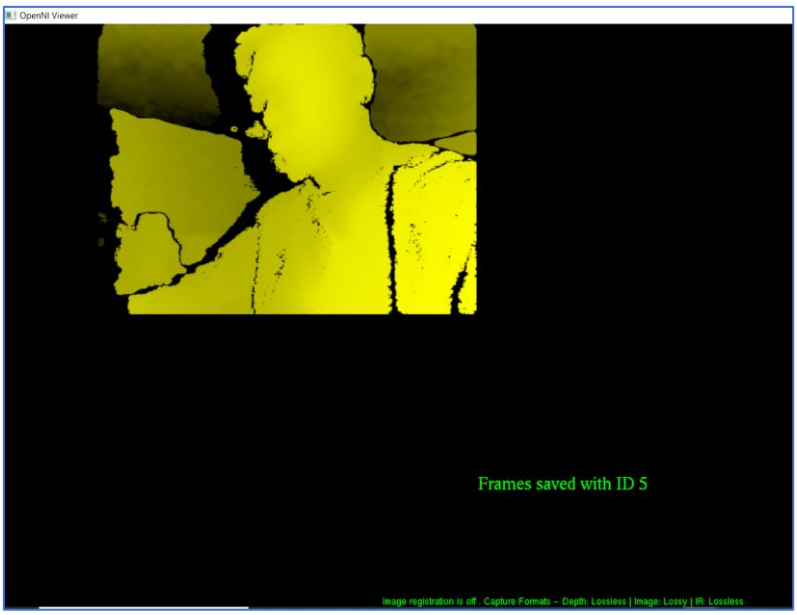
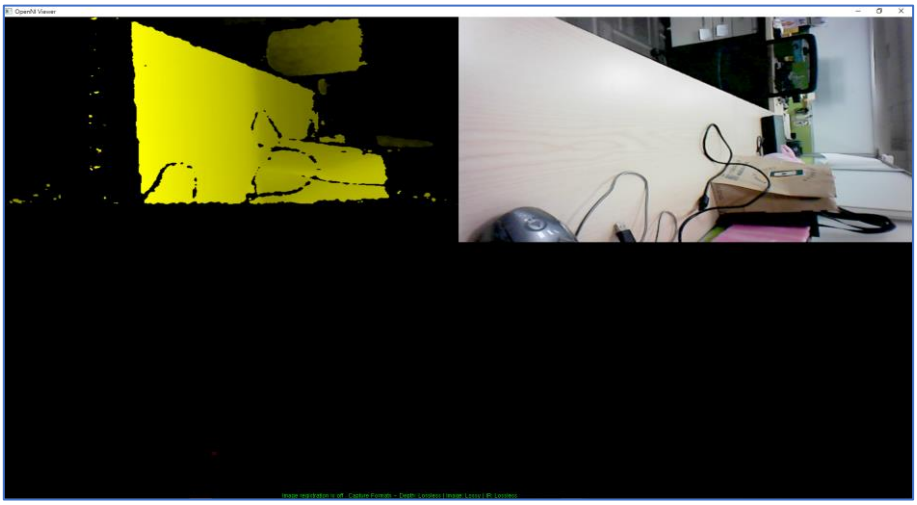
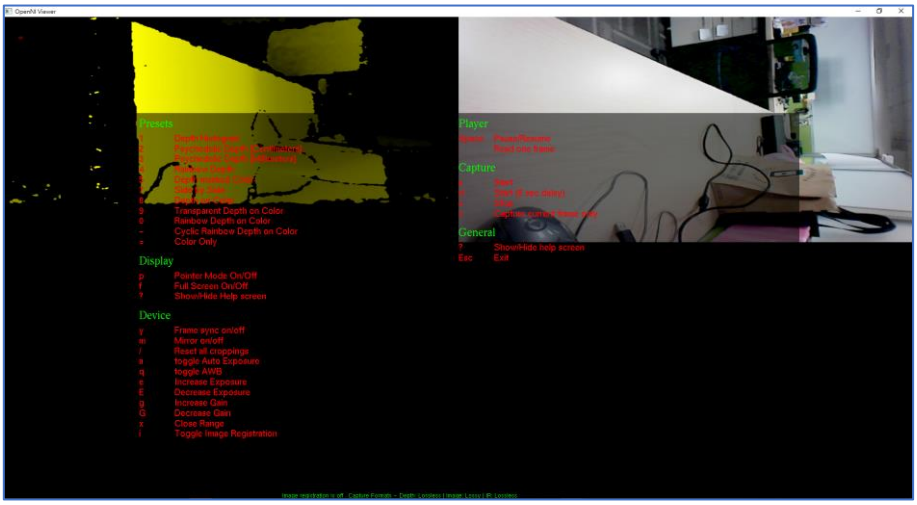
8. Press C to capture the current screen as a raw format file. Go to LIPSedge™ L210 SDK > OpenNI2 > Tools and drag the captured image file to the RawFileViewer.exe to access the raw format file. For details, refer to *Chapter5 5.1-E*.



c. DepthViewer Settings

DepthViewer relies on hotkeys, which are listed in the help menu, to control its functionality and adjustments. By default, the menu in DepthViewer is hidden to provide an unobstructed viewing experience. To access this menu, press Shift + ?.

Default – The help menu hidden

The help menu displayed


The help menu provides the following functionalities:

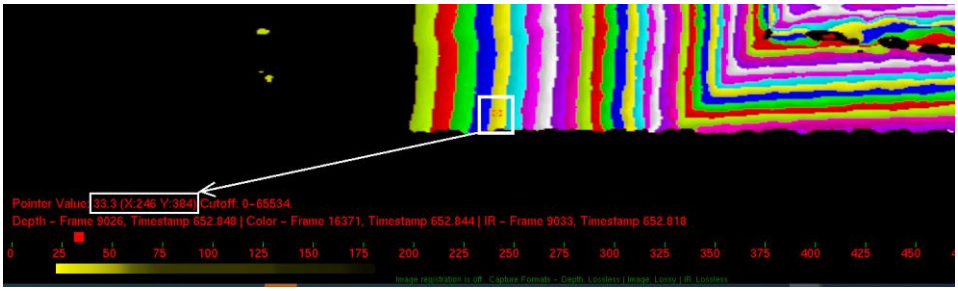
General

- **Show / Hide Help Screen [Shift + ?]**: Shows the **Help Screen** which contains the description of keyboard shortcuts with their functions.
- **Esc**: Exits DepthViewer.+

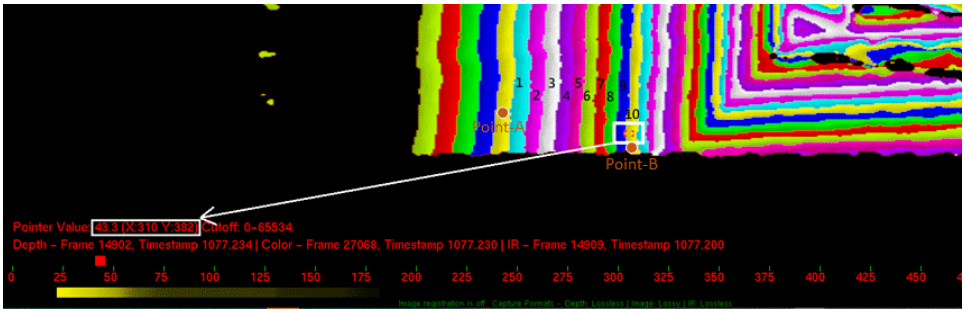
Preset

- **Presets**: Displays the depth/color images in 11 styles. Press the hotkey mentioned by the screen hint to select the image styles. By default, depth/color images are displayed **Side By Side**. In this example, LIPS Corp. demonstrates a way to assess the depth image quality using the psychedelic depth style with the pointer mode [p].
  - **Psychedelic depth [3]**: Visualizes the depth images in a psychedelic style, aiding in the evaluation of image capturing presentation and quality.

    33.3 represents the depth value from the camera lens to the object. (The unit is cm.)

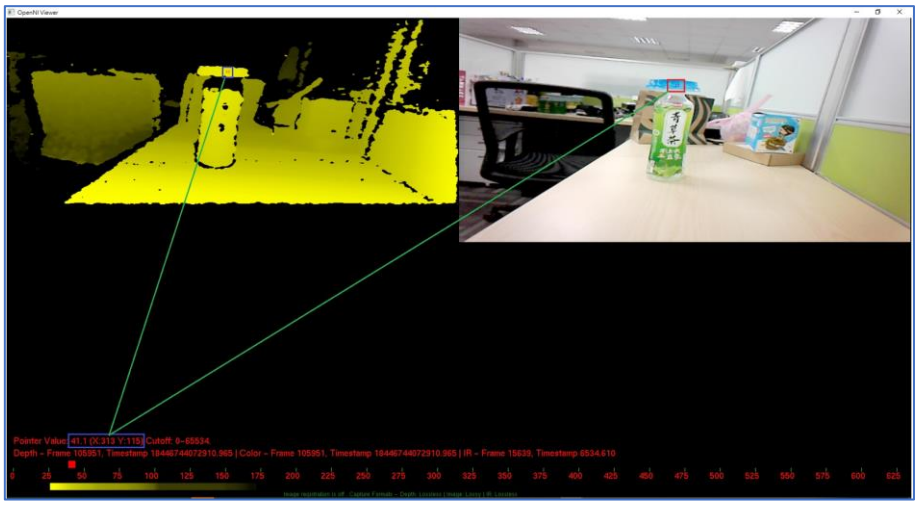
    

    There will be 9 intervals from one color area to the same one-color area, and each interval means 1cm unit. This means the depth value will increase by 10cm from one color area to the same one-color. For example, the depth value will become 43.3 from Point A to Point B.

Display

- **Pointer Mode On / Off [p]**: Enables **Pointer Mode** to show a red dot, which indicates the depth value (unit: mm), representing the distance from the camera lens to the object at a specific point on the image.



- **Full Screen On / Off [f]:** Displays the camera images in full screen or turns the full screen display off.
- **Reset IR histogram [h]:** Clears the IR histogram.
- **Show / hide help screen [shift+?]:** Shows the **Help Screen** which contains the description of keyboard shortcuts with their functions.

Device

- **Depth-Color Frame Sync [y]**: Synchronize the timing between depth/color images.
- **Depth-IR Frame Sync [u]**: Synchronize the timing between depth/color images.
- **Zoom crop on/off [z]:** Enable/disable a cropping mode that magnifies the displayed area, focusing on a specific image region.
- **Mirror All [m]**: Flips the streaming image horizontally for every stream.
- **Reset all croppings [/]**: Reset the cropping area previously assigned.
- **Toggle Auto Exposure [a]**: Automatically adjusts exposure settings for optimal image quality.
- **Toggle AWB [q]**: Enables or disables automatic white balance adjustment.
- **Increase Exposure [e]**: Raises the camera's exposure level for brighter images.
- **Decrease Exposure [E]**: Lowers the camera's exposure level for darker images.

- **Increase Gain [g]**: Enhances image brightness by increasing the camera's gain.
- **Decrease Gain [G]**: Reduces gain to control image brightness.
- **Toggle Close Range [x]**: Toggles between close-range and default focus settings.
- **Toggle Image Registration [i]**: Activates or deactivates image registration.
- **IR Emitter On / Off [t]**: Turns the infrared emitter on or off for depth image capture.

Player

- **Pause / Resume [Space]:** Pauses the playback of the recording or resumes playback.
- **Read one Frame [;]**: This function appears only when the playback is paused. Click to load the playback image of the next frame.

Capture

- **Start [s]**: Assign a path for saving the recording files and recording images.
- **Start (5 sec delay) [d]**: Recording images 5 seconds upon clicking this function.
- **Stop [x]**: Stops recording camera image.
- **Capture current frame only [c]**: Saves the current frame as an image file.

d. IP Address Configuration

This section does NOT apply to **LIPSedge™ L210u / L215u** since a web interface is unavailable for these models.

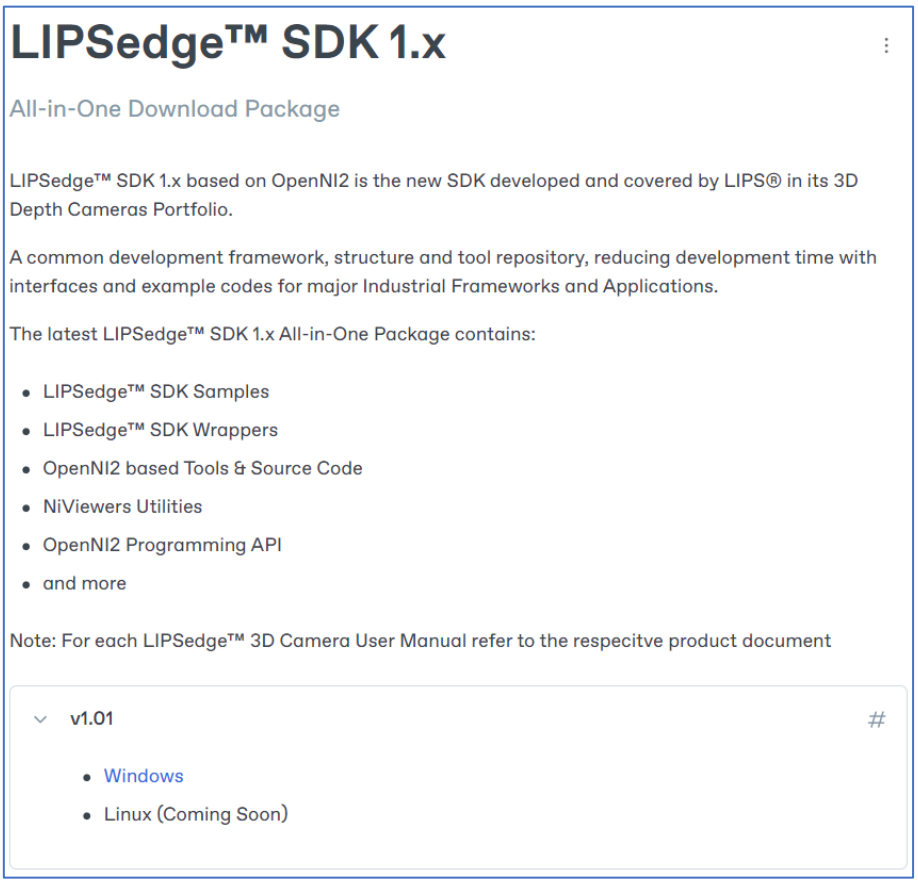
B. Linux / Ubuntu Installation

a. SDK Installation

To access the SDK, visit 3D Depth Camera SDK Free Download | LIPSedge™ SDK (lips-hci.com) and procure the package for the preferred version, framework, and compatible operating system.

1. Visit LIPSedge™ SDK website.

2. Select the installation file based on the compatible operating system and download the installation file.



LIPSedge™ SDK 1.x

All-in-One Download Package

LIPSedge™ SDK 1.x based on OpenNI2 is the new SDK developed and covered by LIPS® in its 3D Depth Cameras Portfolio.

A common development framework, structure and tool repository, reducing development time with interfaces and example codes for major Industrial Frameworks and Applications.

The latest LIPSedge™ SDK 1.x All-in-One Package contains:

- LIPSedge™ SDK Samples
- LIPSedge™ SDK Wrappers
- OpenNI2 based Tools & Source Code
- NiViewers Utilities
- OpenNI2 Programming API
- and more

Note: For each LIPSedge™ 3D Camera User Manual refer to the respecitve product document

v1.01 #

- Windows
- Linux (Coming Soon)

3. Start **terminal** and go to **the installation file's location**.

4. Change the **installation file's access permissions. Version 1.01 is an example;** please key in the version users download.

```
chmod +x LIPSedge-Camera-SDK-Linux-1.01.run
```

```
File  Edit  View  Search  Terminal  Help
lips@lips-Inspiron-7570:~$ chmod +x LIPSedge-Camera-SDK-Linux-1.01.run
lips@lips-Inspiron-7570:~$
```

5. Execute the installation file. Version 1.01 is an example, please key in the version users download.

```
./LIPSedge-Camera-SDK-Linux-1.01.run
```

```
File  Edit  View  Search  Terminal  Help
lips@lips-Inspiron-7570:~$ ./LIPSedge-Camera-SDK-Linux-1.01.run
```

6. The file will be decompressed into model-specific installation files.

```
File  Edit  View  Search  Terminal  Help
lips@lips-Inspiron-7570:~$ ./LIPSedge-Camera-SDK-Linux-1.01.run
Creating directory LIPSedge-Camera-SDK-Linux-1.01
Verifying archive integrity...  100%   MD5 checksums are OK. All good.
Uncompressing LIPSedge Camera SDK Package v1.01 for Linux amd64  100%
LIPSedge SDK All-in-one Package Release Notes

version 1.01 (11/27/2023)
==========================================

This software pacakge contains the following LIPSedge camera SDK:
- LIPSedge DL/M3       version 2.4.4.2_v1.8.2
- LIPSedge AE400/450   version 2.4.4.2_v1.5.1
- LIPSedge AE430/470   version 2.4.4.2_v1.3.1
- LIPSedge L210u/215u version 2.4.4.2_v1.6.0
```

7. Go to **the folder with the installation file for the intended model** and execute the installation file. In this example, LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run is used.

```
chmod +x LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run
```

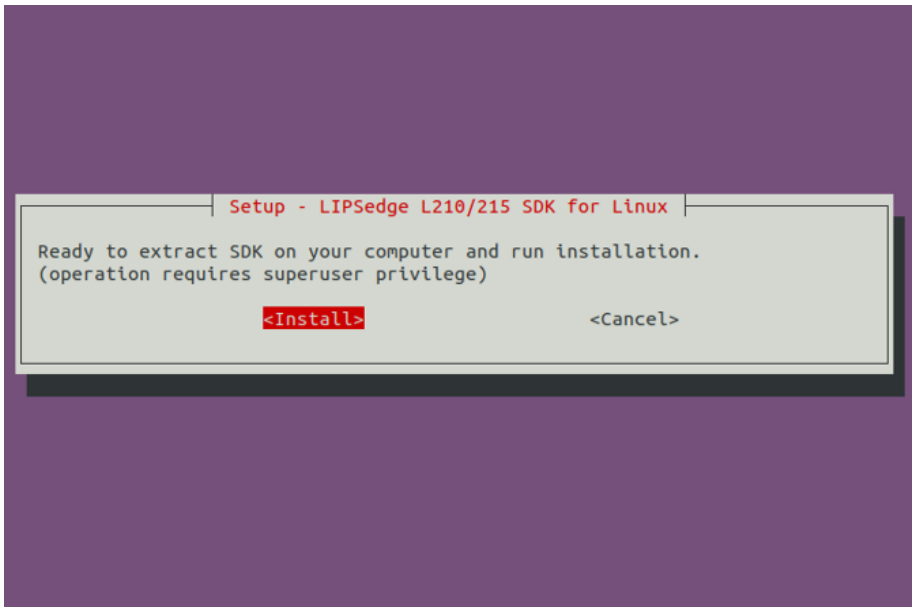
```
./LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run
```

```
lips@lips-Inspiron-7570:~/LIPSedge-Camera-SDK-Linux-1.01$ ./LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run
End-User License Agreement("EULA")
==================================

Last updated: (2017/10/26)

Before you install or otherwise use the software (the "Software") which
is enclosed with this End User License Agreement (this "EULA"), please
read this EULA carefully. This EULA is a legally binding contract
between LIPS Corporation ("LIPS") and you, on behalf of yourself and
the company or another entity on which behalf you are acting ("User").
By installing or otherwise using the Software, User agrees to accept
all terms and conditions of this EULA. If User does not accept this
EULA, you are not allowed to install or otherwise use the Software.

Section 1. Ownership
--------------------
Nothing contained herein shall transfer or be deemed to transfer to
User any title, interest or intellectual property rights relating to
the Software, which shall remain an exclusive property of LIPS and/or
respective third parties including licensor(s) of LIPS.

Section 2. License
------------------
```

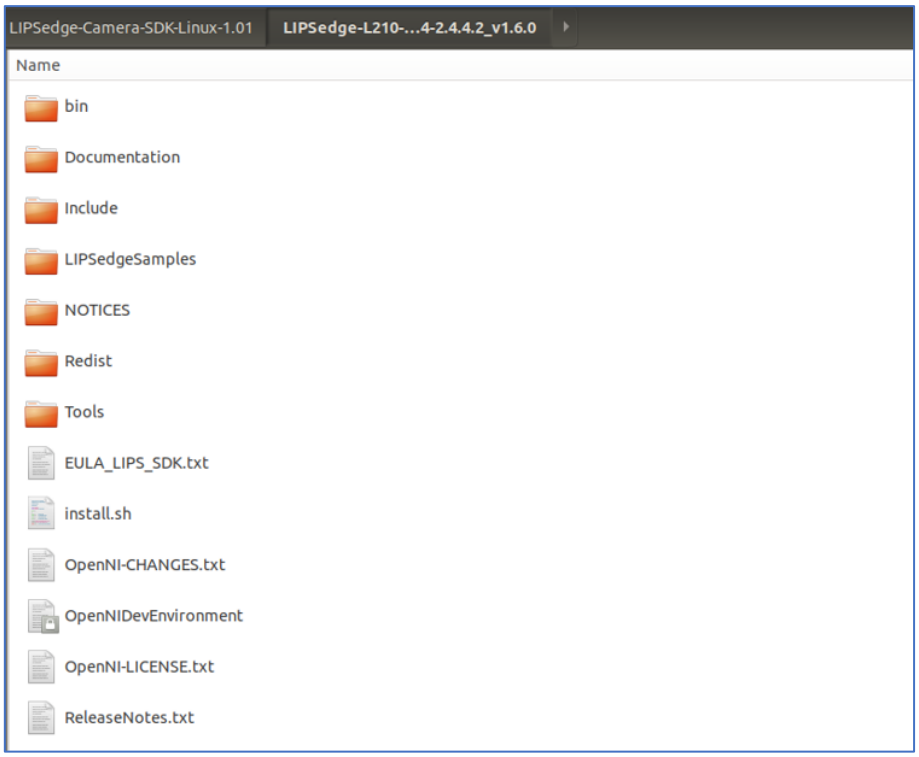
8. Press **Y**.

```
File  Edit  View  Search  Terminal  Help
        in Sections 3 through 6, and 10 shall survive even after this
        EULA is terminated.
    e.  If any part hereof is found void or unenforceable, it will not
        affect the validity of the balance hereof, which shall remain
        valid and enforceable according to its terms.

Section 10. Governing Laws and Jurisdiction
-------------------------------------------
This EULA shall be interpreted and construed under the laws of Taiwan
(R.O.C) without regard to provisions concerning conflicts of law. The
parties hereby submit to jurisdiction of and agree that any legal
proceeding with respect to or arising under this MOU shall be brought
solely in Taiwan Shihlin District Court as court of first stance.

Section 11 General Provisions
-----------------------------
    a.  The failure of a party to enforce any provision hereof shall
        not constitute a waiver of such provision or the right of such
        party to enforce such provision or any other provision.
    b.  This is the entire agreement between LIPS and User relating to
        the Software and it supersedes any prior or contemporaneous
        representations, discussions, undertakings, communications or
        advertising relating to the Software.
Please type y to accept, n otherwise: y
```

9. Select **Install**.



10. Once the installation is successful, the LIPSedge™ SDK and its relevant components are accessible at the **installation location**. In this example, the installation location is ./home. For the folder content, refer to similar content in *Chapter3 3.2-A.*

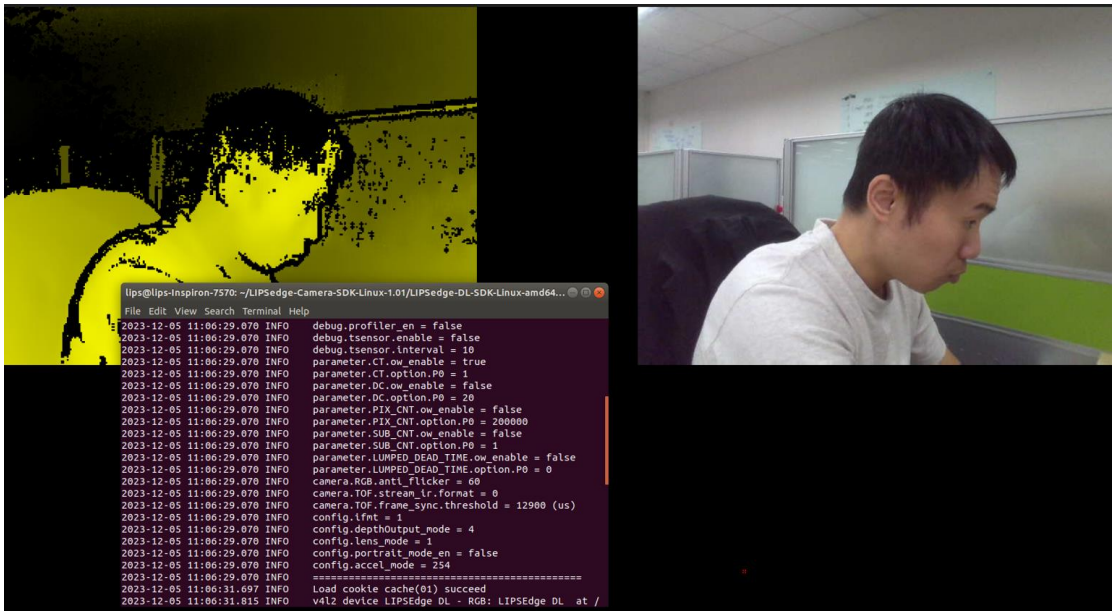


b. Camera Access

LIPSedge™ L210u / L215u SDK (OpenNI-based) utilizes DepthViewer, OpenNI's signature tool, for examining and assessing the quality of depth, RGB, and point cloud images. **Refer to *Chapter3 3.1-A-b* for details.**

1. Go to **LIPSedge-Camera-SDK-Linux** > **LIPSedge-[Model]-SDK** and use the command "**./DepthViewer** "to start DepthViewer.

2. Upon activation, a **Status Window** and a **Viewer Window** both pop



c. DepthViewer Settings

DepthViewer relies on hotkeys listed in the help menu to control its functionality and adjustments. The hotkey functions for the DepthViewer of the Linux version are identical to its Windows counterpart. Refer to *Chapter3 3.1-A-c* for details.

d. IP Address Configuration

This section does NOT apply to **LIPSedge™ L210u / L215u** since a web interface is unavailable for these models.

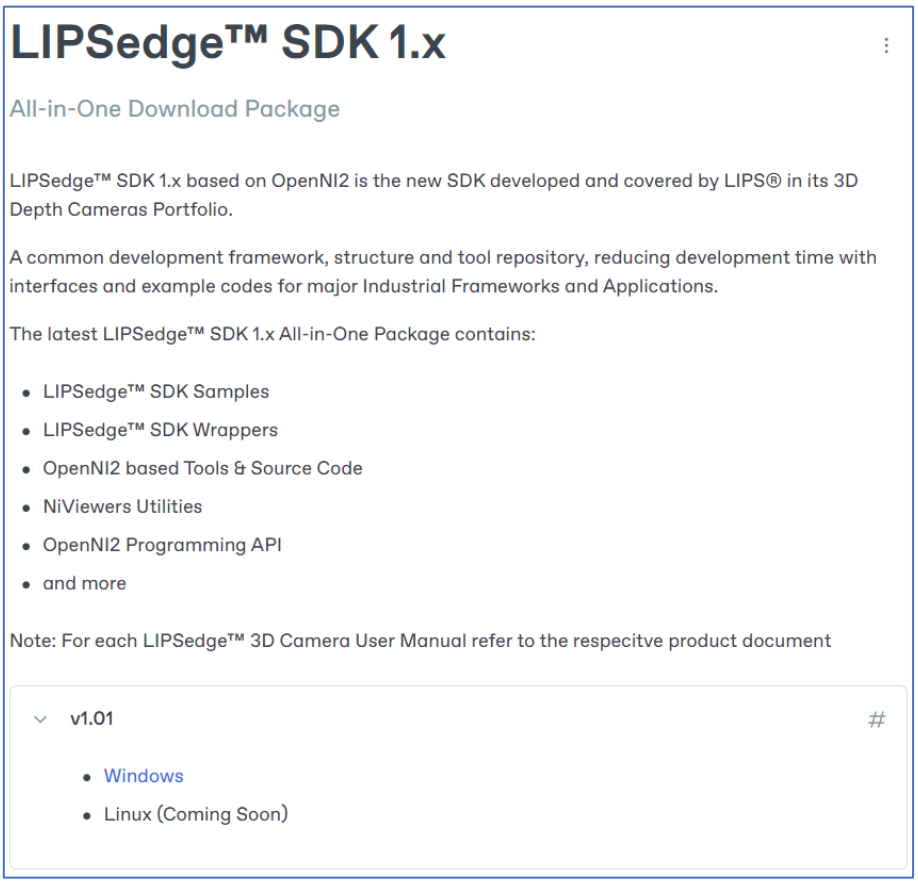
C. NVIDIA Jetson

a. SDK Installation

To access the SDK, visit 3D Depth Camera SDK Free Download | LIPSedge™ SDK (lips-hci.com) and procure the package for the preferred version, framework, and compatible operating system.

1. Visit LIPSedge™ SDK website.

2. Select the installation file based on the compatible operating system and download the installation file.



LIPSedge™ SDK 1.x

All-in-One Download Package

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A common development framework, structure and tool repository, reducing development time with interfaces and example codes for major Industrial Frameworks and Applications.

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- OpenNI2 based Tools & Source Code
- NiViewers Utilities
- OpenNI2 Programming API
- and more

Note: For each LIPSedge™ 3D Camera User Manual refer to the respecitve product document

v1.01 #

- Windows
- Linux (Coming Soon)

3. Start **terminal** and go to **the installation file's location.**

4. Change the **installation file's access permissions. Version 1.01 is an example;** please key in the version users download.

```
chmod +x LIPSedge-Camera-SDK-Linux-1.01.run
```

```
File  Edit  View  Search  Terminal  Help
lips@lips-Inspiron-7570:~$ chmod +x LIPSedge-Camera-SDK-Linux-1.01.run
lips@lips-Inspiron-7570:~$
```

5. Execute the installation file. Version 1.01 is an example, please key in the version users download.

```
./LIPSedge-Camera-SDK-Linux-1.01.run
```

```
File  Edit  View  Search  Terminal  Help
lips@lips-Inspiron-7570:~$ ./LIPSedge-Camera-SDK-Linux-1.01.run
```

6. The file will be decompressed into model-specific installation files.

```
File  Edit  View  Search  Terminal  Help
lips@lips-Inspiron-7570:~$ ./LIPSedge-Camera-SDK-Linux-1.01.run
Creating directory LIPSedge-Camera-SDK-Linux-1.01
Verifying archive integrity...  100%   MD5 checksums are OK. All good.
Uncompressing LIPSedge Camera SDK Package v1.01 for Linux amd64  100%
LIPSedge SDK All-in-one Package Release Notes

version 1.01 (11/27/2023)
=============================================

This software pacakge contains the following LIPSedge camera SDK:
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- LIPSedge AE400/450   version 2.4.4.2_v1.5.1
- LIPSedge AE430/470   version 2.4.4.2_v1.3.1
- LIPSedge L210u/215u version 2.4.4.2_v1.6.0
```

7. Go to **the folder with the installation file for the intended model** and execute the installation file. In this example, LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run is used.

```
chmod +x LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run
```

```
./LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run
```

```
lips@lips-Inspiron-7570:~/LIPSedge-Camera-SDK-Linux-1.01$ ./LIPSedge-L210-SDK-Linux-amd64-2.4.4.2_v1.6.0.xz.run
End-User License Agreement("EULA")
==================================

Last updated: (2017/10/26)

Before you install or otherwise use the software (the "Software") which
is enclosed with this End User License Agreement (this "EULA"), please
read this EULA carefully. This EULA is a legally binding contract
between LIPS Corporation ("LIPS") and you, on behalf of yourself and
the company or another entity on which behalf you are acting ("User").
By installing or otherwise using the Software, User agrees to accept
all terms and conditions of this EULA. If User does not accept this
EULA, you are not allowed to install or otherwise use the Software.

Section 1. Ownership
--------------------
Nothing contained herein shall transfer or be deemed to transfer to
User any title, interest or intellectual property rights relating to
the Software, which shall remain an exclusive property of LIPS and/or
respective third parties including licensor(s) of LIPS.

Section 2. License
```

8. Press **Y**.

```
File  Edit  View  Search  Terminal  Help
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solely in Taiwan Shihlin District Court as court of first stance.

Section 11 General Provisions
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        not constitute a waiver of such provision or the right of such
        party to enforce such provision or any other provision.
    b.  This is the entire agreement between LIPS and User relating to
        the Software and it supersedes any prior or contemporaneous
        representations, discussions, undertakings, communications or
        advertising relating to the Software.
Please type y to accept, n otherwise: y
```
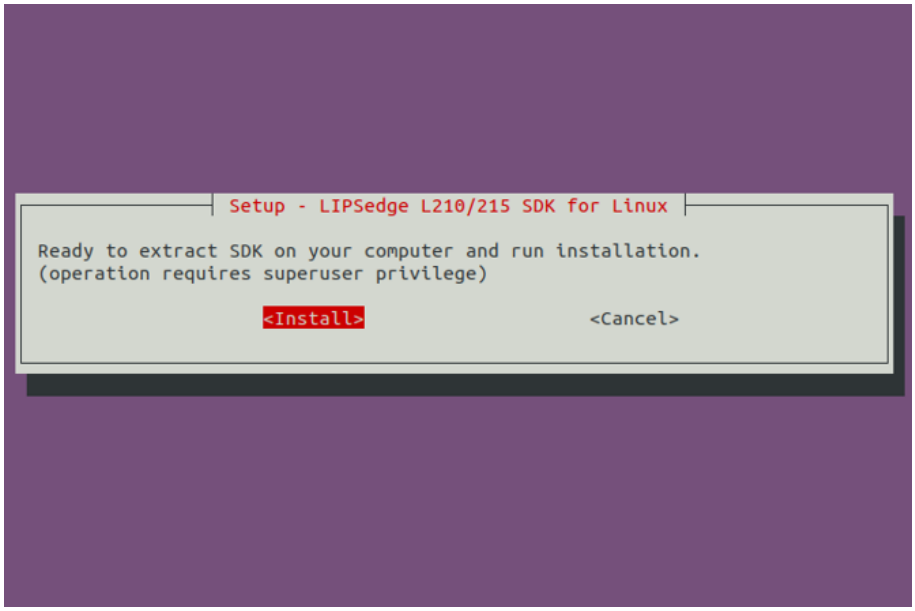
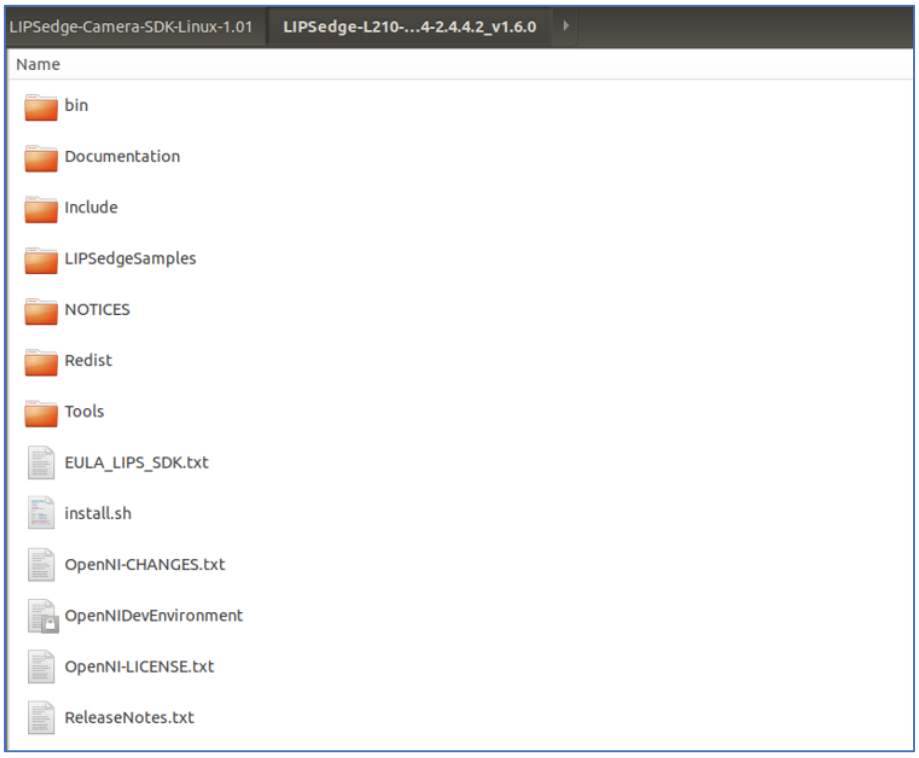

9. Select **Install**.



10. Once the installation is successful, the LIPSedge™ SDK and its relevant components are accessible at the **installation location**. In this example, the installation location is ./home. For the folder content, refer to similar content in *Chapter3 3.2-A.*

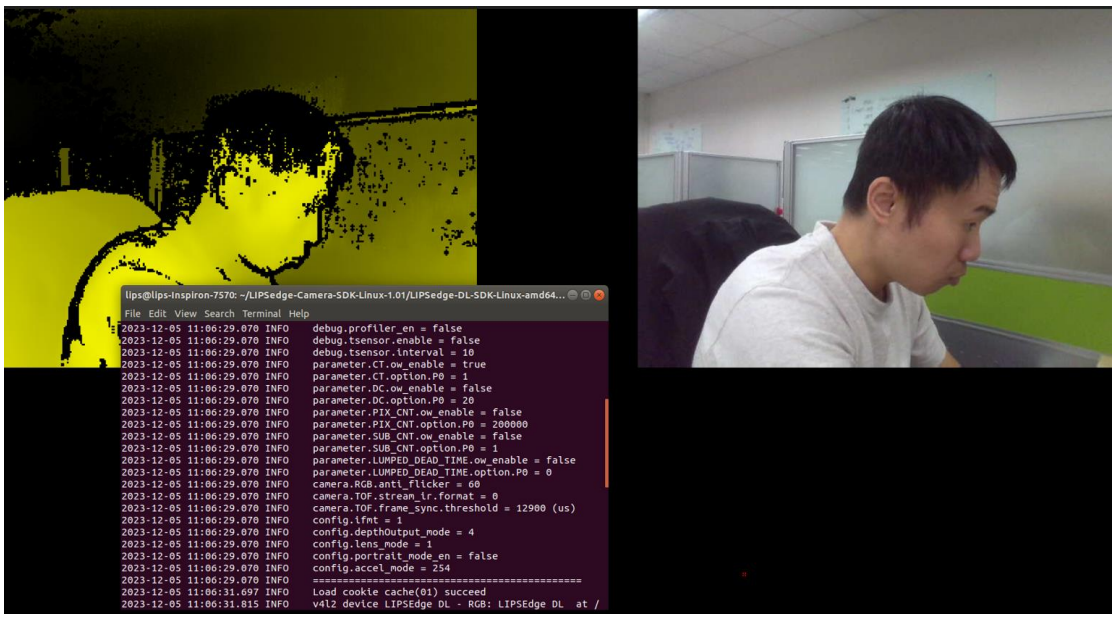


b. Camera Access

LIPSedge™ L210u / L215u utilizes DepthViewer, OpenNI's signature tool, to examine and assess the quality of depth, RGB, and point cloud images. **Refer to *Chapter3 3.1-A-b* for details.**

1. Go to **LIPSedge-Camera-SDK-Linux** > **LIPSedge-[Model]-SDK** and use the command "**./DepthViewer** "to start DepthViewer.

2. Upon activation, a **Status Window** and a **Viewer Window** both pop



c. DepthViewer Settings

DepthViewer relies on hotkeys listed in the help menu to control its functionality and adjustments. The hotkey functions for the DepthViewer of the Linux version are identical to its Windows counterpart. Refer to *Chapter3 3.1-A-c* for details.

d. IP Address Configuration

This section does NOT apply to **LIPSedge™ L210u / L215u** since a web interface is unavailable for these models.

D. Other OS (by request)

Other OS support by request.

3.2 SDK Knowledge Base

The LIPSedge™ L210u / L215u SDK (OpenNI-based) consists of two main components: OpenNI2 and **other dependent files**. Except for the OpenNI2 folder, all the rest of the files are dependent files.

A. SDK Folder Structure

a. Windows

Name	Type	Date modified
NOTICE	File folder	2023/10/4 2:57 PM
OpenNI2	File folder	2023/10/4 2:57 PM
EULA_LIPS_SDK.txt	Text Document	2023/9/11 7:09 PM
LIPS.ico	Icon	2023/9/11 7:09 PM
ReleaseNotes.txt	Text Document	2023/9/26 5:23 PM
unins000.dat	DAT	2023/10/4 2:57 PM
unins000.exe	Application	2023/10/4 2:57 PM

- **NOTICE**: Dependent libraries necessary for LIPSedge™ L210u / L215u SDK functionality. Do NOT modify or delete the folder content.
- **OpenNI2**: OpenNI-related files essential for camera integration and function control, including system configuration files and example applications.
- **Auxiliary files**: Supplementary information for the LIPSedge™ L210u / L215u SDK.
  - **EULA_LIPS_SDK.txt**: The end user agreement.
  - **LIPS.ico**: LIPS Corp.'s icon, which appears during application operation.
  - **ReleaseNotes**: Announcement of the latest feature.
- **Uninstallation components**: Run unins000.exe for uninstallation. Note that unins000.dat must be present.

[OpenNI2 Folder]

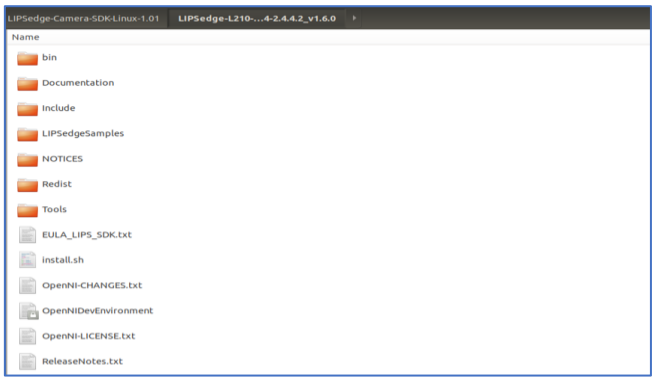
The SDK package for LIPSedge™ L210u / L215u provides a comprehensive set of files and folders to support camera integration and development. These resources include OpenNI C++ API documentation, header files, libraries for compilation, sample applications, system files, and essential information such as licenses and release notes.

Name	Type
Documentation	File folder
Include	File folder
Lib	File folder
LIPSedgeSamples	File folder
Redist	File folder
Tools	File folder
CHANGES.txt	Text Document
LICENSE	File
NOTICE	File
ReleaseNotes.txt	Text Document

- Documentation: Contains essential OpenNI C++ API documentation.
- Include: Contains essential OpenNI C++ headers. These headers can be checked along with the associated library files to facilitate development.
- Lib: Contains library files, specifically OpenNI2.lib, and realsenses2.lib. These libraries are used in conjunction with MS Visual Studio C++ for compilation.
- LIPSedgeSamples: Contains a series of executable tools and the source code of these tools.
- Redist: Contains network configuration files, camera drivers, and system files necessary for image streaming.
- Tools: Holds captured frames from example programs and other camera configuration files.
- CHANGES: The change log for OpenNI.
- LICENSE: Contains information about the licensing terms for the SDK.
- NOTICE: Contains copyright and attribution notices relevant to the OpenNI components.
- ReleaseNotes: The latest system requirements for OpenNI.

b. Linux

In LIPSedge™ L210u / L215u SDK (OpenNI-based) for the Linux system, the folders are structured so that the OpenNI2 and dependency files are **adjacent to the application that requires these external resources**. Go to **./home/LIPSedge-Camera-SDK-Linux/LIPSedge[Model]-SDK** for OpenNI C++ API documentation, header files, libraries for compilation, sample applications, system files, and essential information such as licenses and release notes.



- **bin**: Consists of the USB driver files for the LIPS camera.
- **Documentation**: Contains essential OpenNI C++ API documentation.
- **Include**: Contains essential OpenNI C++ headers. These headers can be checked along with the associated library files to facilitate development.
- **LIPSedgeSamples**: Contains a series of executable tools and the source code of these tools.
- **NOTICE**: Contains copyright and attribution notices relevant to the OpenNI components.
- **Redist**: Contains network configuration files, camera drivers and system files necessary for image streaming.
- **Tools**: Holds captured frames from example programs and other camera configuration files.
- **EULA_LIPS_SDK.txt**: The end user agreement.
- **Install.sh**: In cases where the Linux / Ubuntu SDK installation was aborted, execute install.sh to install the SDK.
- **OpenNI-CHANGES.txt**: The change log for OpenNI.
- **OpenNIDevEnvironment**:
- **OpenNI-LICENSE.txt**: Contains information about the licensing terms for OpenNI.
- **ReleaseNotes**: The latest system requirements for OpenNI.

B. SDK Setting Configuration

In addition to the OpenNI2 settings, the camera profile settings are also available for adjustment in the **LIPSedge-L210.json** file, written in key / value pairs. To modify the SDK settings, edit LIPSedge-L210.json file in the following location:

Location	
Windows	C:/LIPSedge™ DL SDK path/Tools/OpenNI2/Drivers
Linux	C:/LIPSedge™ DL SDK path/Tools/OpenNI2/Drivers

Open **LIPSedge-L210.json** with **administrator privilege**. Users can change the settings for different applications. Below is the interface of LIPSedge-L210.json.

```
"debug": {
    "logLevel": "verbose",
    "defaultRes": "VGA",
    "showFPS": {
        "enable": false,
        "interval": 10
    }
},
"config": {
    "version": "1.6.3"
},
"post-processing":
{
    "bilateral-filter":
    {
        "enable": true,
        "debug": false,
        "sigma-spatial": 40,
        "sigma-spatial_help": "scope 1 to 100",
        "sigma-range": 21,
        "sigma-range_help": "scope 1 to 100"
    }
}
```

a. Debug

- Loglevel:

Do NOT change this parameter.

```
"logLevel": "verbose",
```

- defaultRes:

  Adjusts the camera resolution. 2 parameters are available: **VGA (640p) / HD (1080p)**.

```
"defaultRes": "VGA",
```

- ShowFPS:

  Adjust the value (**True / False**) for the **enable** parameter to display FPS on-screen. By default, the parameter is set as false.

```
"showFPS": {
    "enable": false,
```

- interval:

  Show the current FPS on the screen at a regular interval (seconds). For example, 10 represents displaying the FPS for every 10 seconds.

```
"showFPS": {
    "enable": false,
    "interval": 10
```

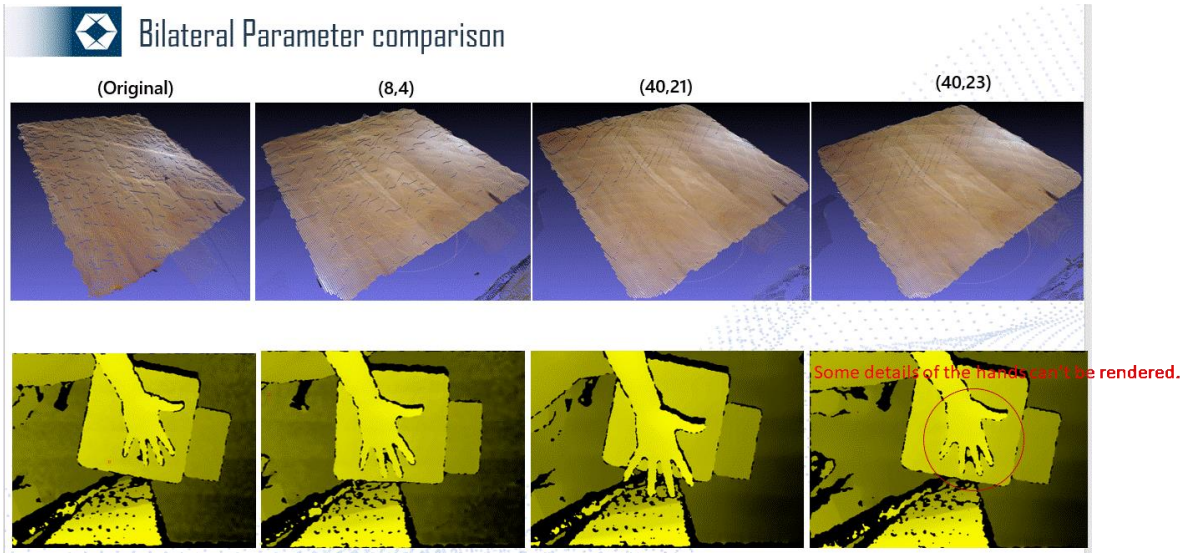
b. Post-Processing.

- Enable:
  If users key in “True”, then the bilateral filter will be turned on, and the parameters will be implemented by the default setting below.

```
"sigma-spatial": 40,
"sigma-spatial_help": "scope 1 to 100",
"sigma-range": 21,
"sigma-range_help": "scope 1 to 100"
```

  If users key in “False”, the bilateral filter will be turned off.

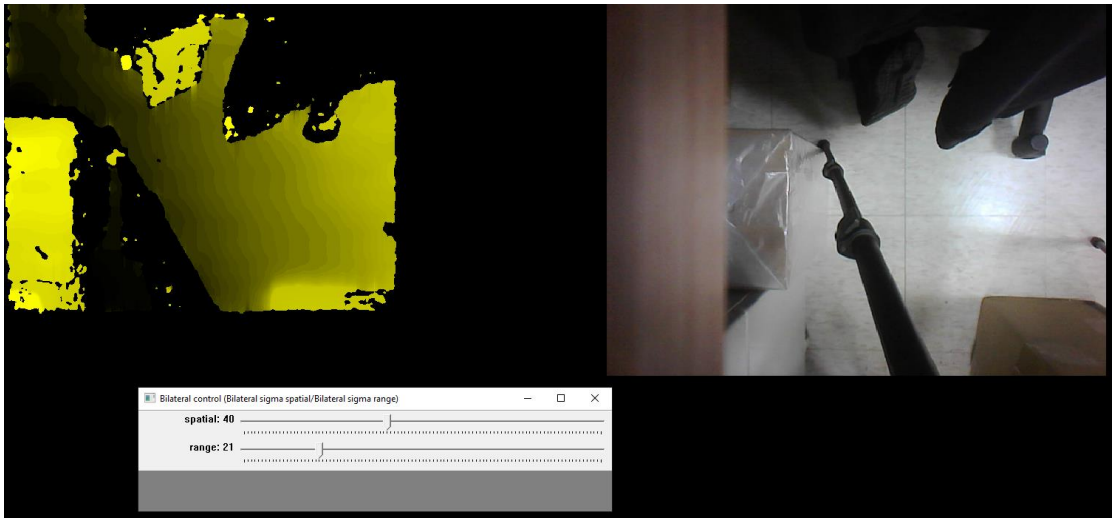
As per the report below, the default setting(40,21) can denoise better than the original setting (which means turning off the bilateral filter).



- **Sigma Spatial:** filter kernel size, increasing it can smooth larger features.
- **Sigma Range:** minimum edge amplitude to be considered,
  increasing it, the bilateral filter gradually approximates Gaussian blur.

- Debug:

If users key in “True” in enable & debug, then the UI of bilateral-filter will be turned on as below; users can see the result of the image by adjusting the parameters.



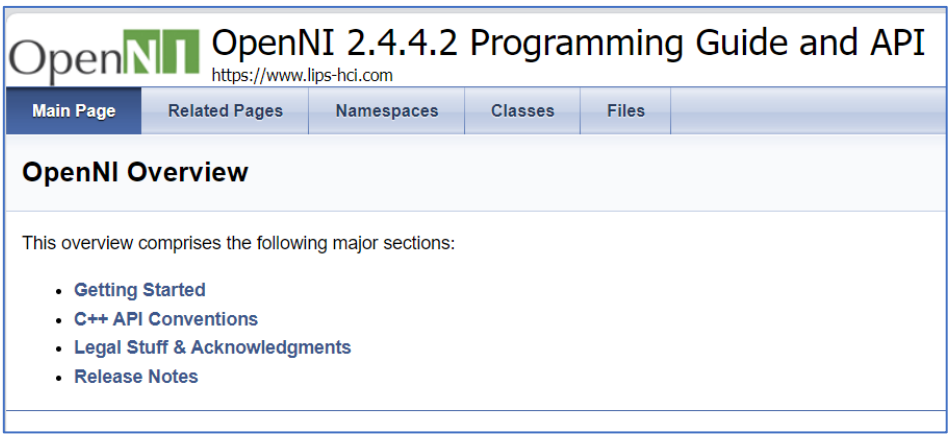
If users key in “False”, then the UI of the bilateral filter will be turned off.

C. Camera Web Interface

This section does NOT apply to **LIPSedge™ L210u / L215u** since a web interface is unavailable for these models.

4. APIs

LIPSedge™ L210u / L215u SDK is integrated with OpenNI APIs. The key strength of this robust framework lies in its effortless access to the camera's internal parameters, empowering developers to delve deep into raw data and tailor applications to specific functionalities. Developers can access the programming guide for the API at: /LIPSedge[Model]-SDK/Documentation.



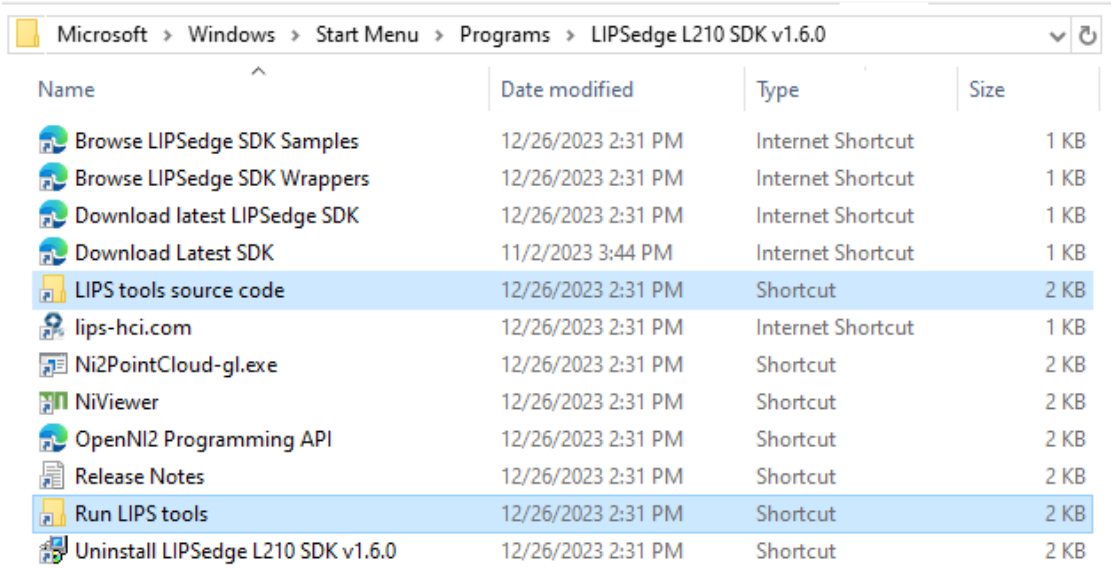
1. Main Page: This tab provides a clear overview of the OpenNI API, including a quick installation guide, introduction of the coding conventions, legal documents, and the release notes.
2. Related Pages: This tab provides a selection of the information that is related to OpenNI.
3. Namespaces: This tab provides users with a hierarchical structure of namespaces that categorize the specific functionalities of the API.
4. Classes: This tab dives deeper into each individual class within the OpenNI API, outlining its properties and methods to enable developers to fully understand and utilize each element.
5. Files: This tab provides a description of the source code files that make up the OpenNI API.

5. Tools

LIPSedge™ L210u / L215u SDK provides a series of executable tools and their source code essential for visualization, data capture, camera parameters acquisition et cetera to facilitate the development of applications and solutions for LIPSedge™ L210u / L215u cameras. These tools and their system components are placed in the following folder:

- Run LIPS tools Source code: Contains a series of executable tools; users can refer to *Chapter 5.1 Samples codes* for reference.

- LIPS tools source code: Contains a series of the source code of these tools. Users can refer to *Chapter 5.2 Compilation* for reference. These codes can also be downloaded from the link below for the latest updates: https://github.com/lips-hci/LIPSedge-sdk-samples.

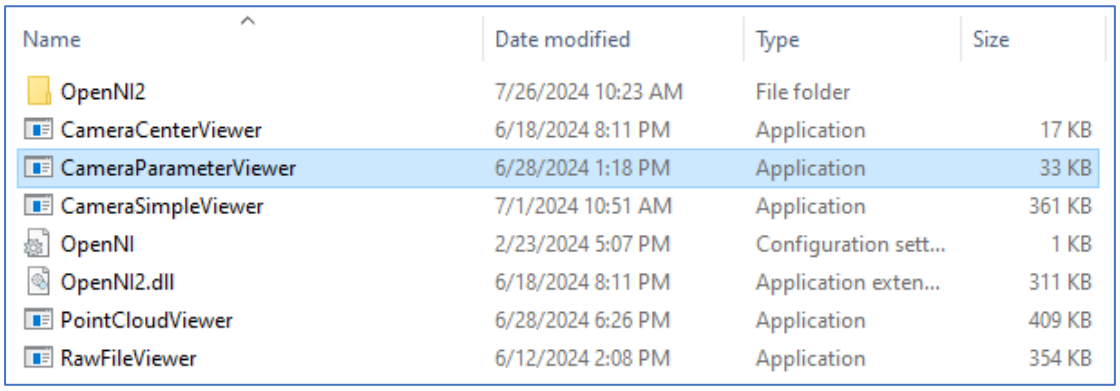


5.1 Sample codes

A. CameraParameterViewer

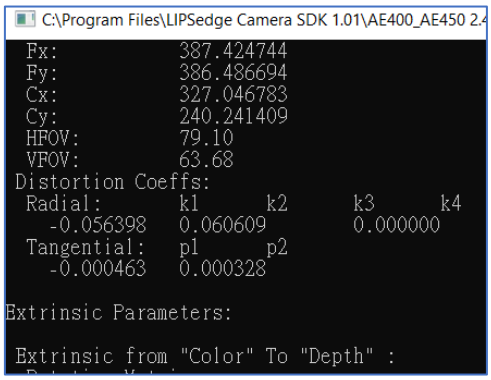
A simple OpenNI2 tool to query supported video modes and list camera intrinsic/extrinsic parameters.

1. In **Bin**, start **Cameraparameter.exe**.



Name	Date modified	Type	Size
OpenNI2	7/26/2024 10:23 AM	File folder	
CameraCenterViewer	6/18/2024 8:11 PM	Application	17 KB
CameraParameterViewer	6/28/2024 1:18 PM	Application	33 KB
CameraSimpleViewer	7/1/2024 10:51 AM	Application	361 KB
OpenNI	2/23/2024 5:07 PM	Configuration sett...	1 KB
OpenNI2.dll	6/18/2024 8:11 PM	Application exten...	311 KB
PointCloudViewer	6/28/2024 6:26 PM	Application	409 KB
RawFileViewer	6/12/2024 2:08 PM	Application	354 KB

2. Provides detailed camera parameters for depth, IR, and color cameras.

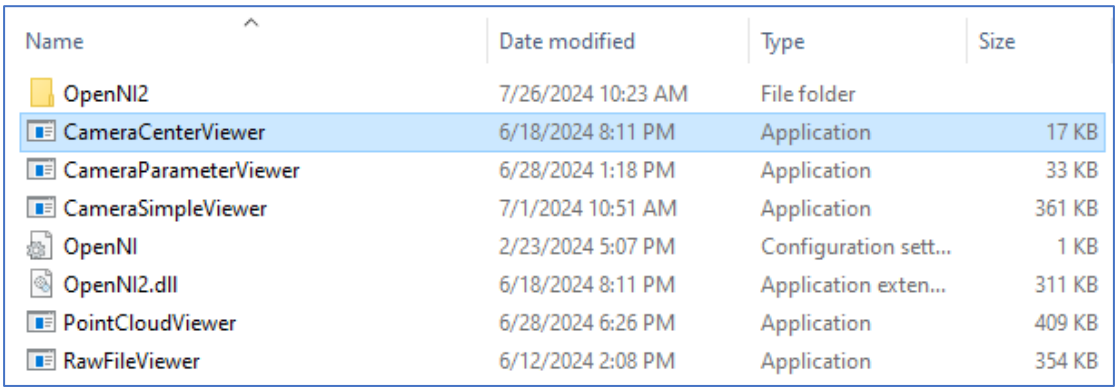


```
C:\Program Files\LIPSedge Camera SDK 1.01\AE400_AE450 2.4
 Fx:            387.424744
 Fy:            386.486694
 Cx:            327.046783
 Cy:            240.241409
 HFOV:          79.10
 VFOV:          63.68
Distortion Coeffs:
 Radial:        k1       k2       k3       k4
   -0.056398    0.060609         0.000000
 Tangential:    p1       p2
   -0.000463    0.000328

Extrinsic Parameters:

 Extrinsic from "Color" To "Depth" :
```

B. Ni2CenterRead

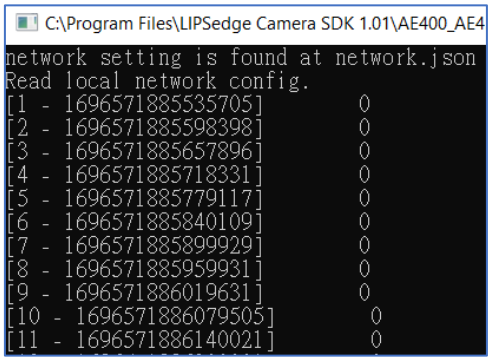
A simple OpenNI2 console program to show depth value at center point of frame.

1. In **Bin**, click **CameraCenterViewer.**



Name	Date modified	Type	Size
OpenNI2	7/26/2024 10:23 AM	File folder	
CameraCenterViewer	6/18/2024 8:11 PM	Application	17 KB
CameraParameterViewer	6/28/2024 1:18 PM	Application	33 KB
CameraSimpleViewer	7/1/2024 10:51 AM	Application	361 KB
OpenNI	2/23/2024 5:07 PM	Configuration sett...	1 KB
OpenNI2.dll	6/18/2024 8:11 PM	Application exten...	311 KB
PointCloudViewer	6/28/2024 6:26 PM	Application	409 KB
RawFileViewer	6/12/2024 2:08 PM	Application	354 KB

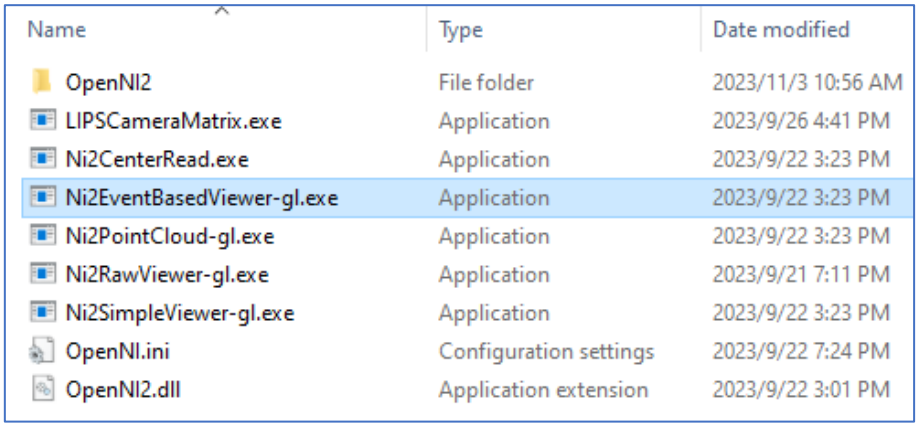
2. A simple OpenNI2 console program to show depth value at center point of frame.



C. Ni2EventBasedViewer-gl (Coming soon)

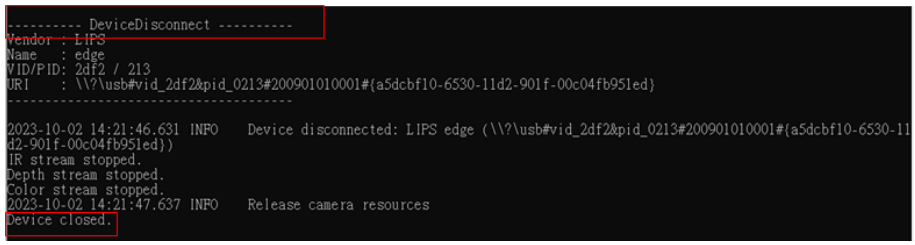
OpenNI2 device management by event-driven design.

1. In **Bin**, click **Ni2EventBasedViewer-gl.**



Name	Type	Date modified
OpenNI2	File folder	2023/11/3 10:56 AM
LIPSCameraMatrix.exe	Application	2023/9/26 4:41 PM
Ni2CenterRead.exe	Application	2023/9/22 3:23 PM
Ni2EventBasedViewer-gl.exe	Application	2023/9/22 3:23 PM
Ni2PointCloud-gl.exe	Application	2023/9/22 3:23 PM
Ni2RawViewer-gl.exe	Application	2023/9/21 7:11 PM
Ni2SimpleViewer-gl.exe	Application	2023/9/22 3:23 PM
OpenNI.ini	Configuration settings	2023/9/22 7:24 PM
OpenNI2.dll	Application extension	2023/9/22 3:01 PM

2. The Ni2EventBasedViewer-gl provides notifications when the device becomes disconnected.



3. When users reconnect the device, the system promptly re-establishes the connection, granting access to streaming for Depth, IR, and RGB data once again.

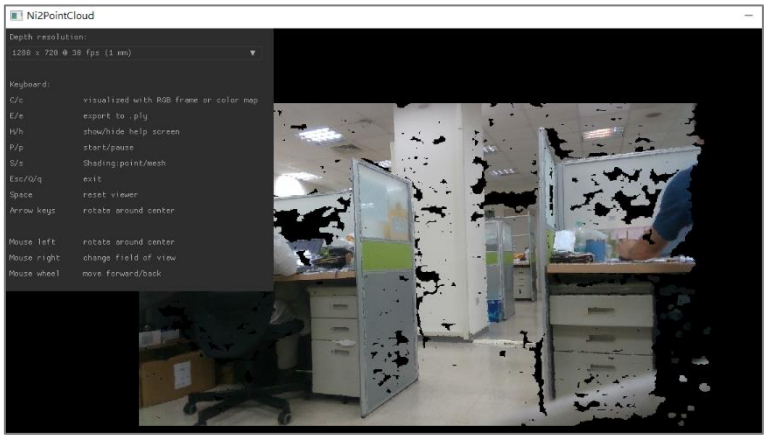
D. PointCloudViewer

Includes another OpenNI2 sample for 3D point cloud visualization through OpenGL.

1. In **Bin**, click **PointCloudViewer**

Name	Date modified	Type	Size
OpenNI2	7/26/2024 10:23 AM	File folder	
CameraCenterViewer	6/18/2024 8:11 PM	Application	17 KB
CameraParameterViewer	6/28/2024 1:18 PM	Application	33 KB
CameraSimpleViewer	7/1/2024 10:51 AM	Application	361 KB
OpenNI	2/23/2024 5:07 PM	Configuration sett...	1 KB
OpenNI2.dll	6/18/2024 8:11 PM	Application exten...	311 KB
PointCloudViewer	6/28/2024 6:26 PM	Application	409 KB
RawFileViewer	6/12/2024 2:08 PM	Application	354 KB

2. A simple viewer application using OpenGL to display point-cloud frames.



E. RawFileViewer

A tool for reading OpenNI2 frames captured in raw data format. To achieve this:

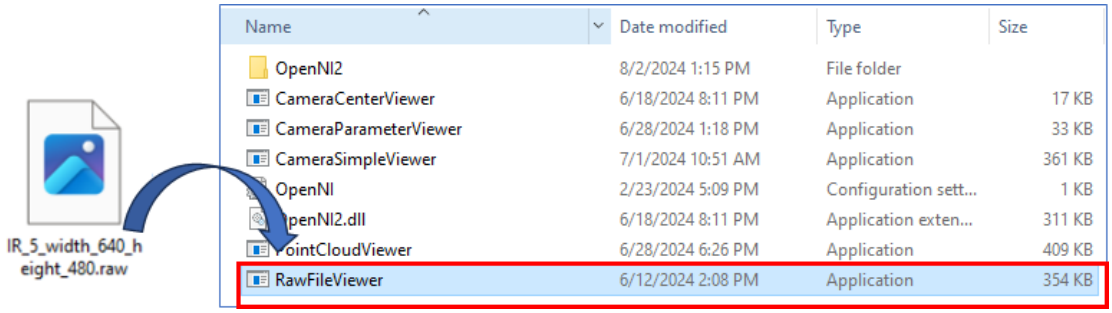
Name	Date modified	Type	Size
OpenNI2	7/26/2024 10:23 AM	File folder	
CameraCenterViewer	6/18/2024 8:11 PM	Application	17 KB
CameraParameterViewer	6/28/2024 1:18 PM	Application	33 KB
CameraSimpleViewer	7/1/2024 10:51 AM	Application	361 KB
OpenNI	2/23/2024 5:07 PM	Configuration sett...	1 KB
OpenNI2.dll	6/18/2024 8:11 PM	Application exten...	311 KB
PointCloudViewer	6/28/2024 6:26 PM	Application	409 KB
RawFileViewer	6/12/2024 2:08 PM	Application	354 KB

1. Start DepthViewer. Make sure the camera is properly configured and connected.

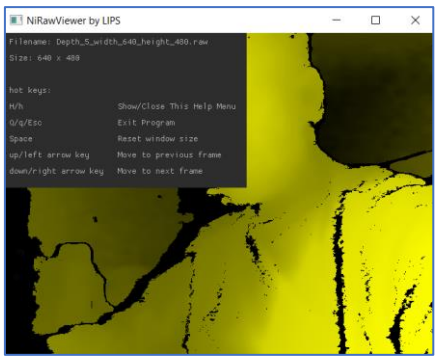
2. Press C and the current image will be captured.

3. Go to **LIPSedge™ L210 SDK > OpenNI2 > Tools.**

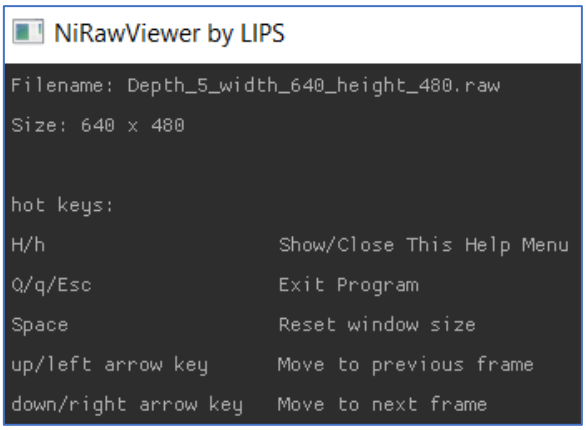
4. Drag a raw image file to the RawFileViewer-gl.



5. The raw image file will be displayed.



6. Users can navigate through the image with the following hotkeys:
   - **H / h**: Displays / turn off the help menu.
   - **Q / q / Esc**: Exit NiRawViewer-gl.
   - **Space**: Reset adjusted window size to its original resolution.
   - **Up / Left arrow key**: Displays the previous frame.
   - **Down / Right arrow key**: Displays the next frame.



F. CameraSimpleViewer

A simplified version of DepthViewer using OpenCV to display Depth/IR/Color frames.

3. In **Bin**, click **CameraSimpleViewer**

Name	Date modified	Type	Size
OpenNI2	7/26/2024 10:23 AM	File folder	
CameraCenterViewer	6/18/2024 8:11 PM	Application	17 KB
CameraParameterViewer	6/28/2024 1:18 PM	Application	33 KB
CameraSimpleViewer	7/1/2024 10:51 AM	Application	361 KB
OpenNI	2/23/2024 5:07 PM	Configuration sett...	1 KB
OpenNI2.dll	6/18/2024 8:11 PM	Application exten...	311 KB
PointCloudViewer	6/28/2024 6:26 PM	Application	409 KB
RawFileViewer	6/12/2024 2:08 PM	Application	354 KB

4. A simplified version of DepthViewer using OpenCV to display Depth/IR/Color frames.

5.2 Compilation

Examples application's source code: The source code of the executable tools in LIPSedgeSamples. Refer to *Chapter5 5.1 Samples codes*. In cases where the source code is corrupted, LIPS Corp. provides the source code on LIPS Corp.'s GitHub.

Application name	Source code
LIPSCameraMatrix	LIPSedge-sdk-samples/LIPSCameraMatrix at main · lips-hci/LIPSedge-sdk-samples · GitHub
Ni2CenterRead	LIPSedge-sdk-samples/Ni2CenterRead at main · lips-hci/LIPSedge-sdk-samples · GitHub
Ni2EventBasedViewer-gl	LIPSedge-sdk-samples/Ni2EventBasedViewer-gl at main · lips-hci/LIPSedge-sdk-samples · GitHub
Ni2PointCloid-gl	LIPSedge-sdk-samples/Ni2PointCloud-gl at main · lips-hci/LIPSedge-sdk-samples · GitHub
Ni2RawViewer-gl:	LIPSedge-sdk-samples/Ni2RawViewer-gl at main · lips-hci/LIPSedge-sdk-samples · GitHub
Ni2SimpleViewer-cv	LIPSedge-sdk-samples/Ni2SimpleViewer-cv at main · lips-hci/LIPSedge-sdk-samples · GitHub
Ni2SimpleViewer-gl:	LIPSedge-sdk-samples/Ni2SimpleViewer-gl at main · lips-hci/LIPSedge-sdk-samples · GitHub

6. Supported Language & Wrappers

LIPSedge™ L210u / L215u SDK (OpenNI-based) supports various programming Languages/ Libraries/ Frameworks to enhance the development project. These programming language-based SDK and libraries/ frameworks are available at LIPS-HCI · GitHub.

Type	Name	Description	GitHub Link
Language	Python3	Support Python3	python3 · GitHub
	.NET	Support for .NET languages with examples in C#	.NET GitHub
	C#	Support C#	Coming Soon.
	Java	Support Java	Coming Soon.
Library	OpenCV	Integration with OpenCV computer-vision library	OpenCV GitHub
	PCL	Integration with Point Cloud Library	Coming Soon.
Framework	ROS	Integration with Robot Operating System (ROS)	Coming Soon.
	ROS2	Integration with Robot Operating System 2 (ROS 2)	Coming Soon.
	ISAAC	Support NV Isaac robot platform as depth camera data provider	Coming Soon.
	Halcon	Integration with HALCON development environment	Coming Soon.

7. Application & Middleware Supports

In addition to the LIPSedge™ L210u / L215u, LIPS Corp. also offers a diverse range of advanced 3D cameras and comprehensive solutions that cater to various imaging and sensing requirements. For details, refer to the following chapter:

7.1 Applications

LIPS Corp. offers a wide range of advanced 3D camera kits. These offerings encompass various technologies to cater to diverse imaging and sensing requirements. For details, refer to. https://www.lips-hci.com/.

- LIPSMetric™
- LIPSense™

7.2 Middleware

LIPS Corp. offers exclusive middleware solutions. These solutions encompass various technologies to cater to diverse imaging and sensing requirements. For details, refer to. https://www.lips-hci.com/.

- LIPSMetric™
- LIPSense™

8. SDK Tutorial

LIPS Corp. provides extensive support for a variety of SDKs. LIPS Corp.'s GitBook tutorial offers practical examples and applications with hands-on guidance based on the following platforms:

Name	Description
C++	https://lips-hci.gitbook.io/lips-developer-documentation/sdk-code-samples-and-languages-wrappers/openni2/overview
Python	https://lips-hci.gitbook.io/lips-developer-documentation/sdk-code-samples-and-languages-wrappers/openni2/overview-1
Java	https://lips-hci.gitbook.io/lips-developer-documentation/sdk-code-samples-and-languages-wrappers/openni2/overview-2
OpenCV	https://lips-hci.gitbook.io/lips-developer-documentation/sdk-code-samples-and-languages-wrappers/openni2/overview-3
GenICam	https://lips-hci.gitbook.io/lips-developer-documentation/sdk-code-samples-and-languages-wrappers/openni2/genicam
ROS	https://lips-hci.gitbook.io/lips-developer-documentation/sdk-code-samples-and-languages-wrappers/openni2/overview-4

9. Appendix (Private by owner)

a. Regulatory Compliance Notice

FCC Compliance

Note: This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference when the equipment is operated in a commercial environment. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instruction manual, may cause harmful interference to radio communications. Operation of this equipment in a residential area is likely to cause harmful interference in which case the user will be required to correct the interference at his own expense.

Caution: Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

FCC Label Notice

This device complies with part 15 of the FCC Rules. Operation is subject to the following two conditions:

(1) This device may not cause harmful interference, and

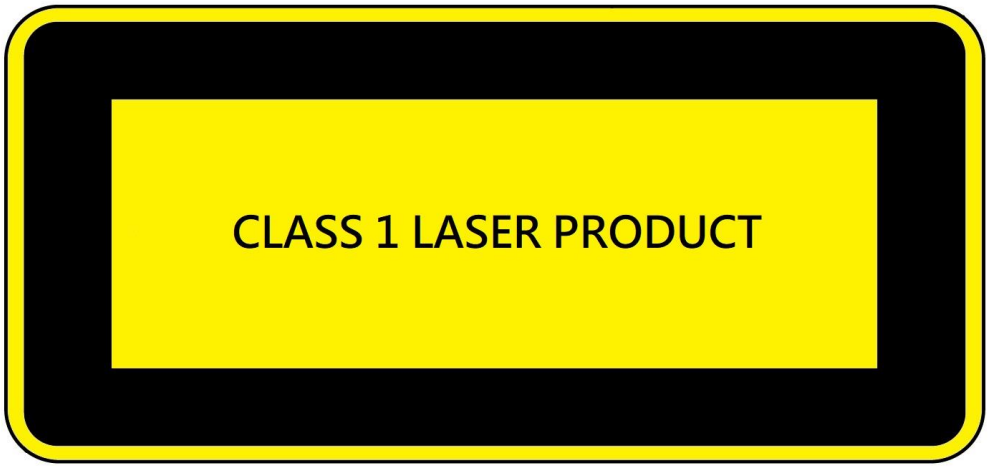
(2) this device must accept any interference received, including interference that may cause undesired operation.

IEC/ EN 60825 + FDA Laser Safety

CLASS 1 LASER PRODUCT

COSUMER LASER PRODUCT

EN 50689:2021



This device complies with FDA performance standards for laser products except for conformance with IEC 60825-1 Ed. 3., as described in Laser Notice No. 56, dated May 8, 2019.

CE Compliance

Warning:

This is a class A product. In a domestic environment, this product may cause radio interference in
which case the user may be required to take adequate measures.

RoHS Compliance

All lead-free products offered by the company comply with the requirements of the European law on the Restriction of Hazardous Substances (RoHS) directive, which means our manufacture processes and products are strictly “lead-free” and without the hazardous substances cited in the directive.

LIPS
Sense The World

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